

ABSTRACT

The sugar and distillery industries generate substantial quantities of organic waste, namely press mud (PM) and raw spent wash (RSW), which pose significant environmental challenges due to their high organic load and complex chemical composition. This study investigates the biochemical methane potential (BMP) of press mud and raw spent wash, individually and in co-digestion, with the objective of evaluating their feasibility as substrates for enhanced biogas production through anaerobic digestion. The substrates were characterized for key physicochemical parameters including moisture content, total solids (TS), volatile solids (VS), chemical oxygen demand (COD), pH, total dissolved solids (TDS), and total suspended solids (TSS). Press mud exhibited a moisture content of 71.59%, total solids of 28.42%, and volatile solids of 20.2%, while raw spent wash recorded a high COD of 140,000 mg/L and a volatile solids fraction of 75.29% of TS, confirming the strong biodegradability of both substrates. BMP experiments were conducted using the Automatic Methane Potential Test System (AMPTS-III) at a mesophilic temperature of 38°C, evaluating mono-digestion of PM and RSW alongside co-digestion at mixing ratios of PM:RSW (1:2) and PM: RSW (1:3). The results demonstrated that press mud, as a sole substrate, achieved a BMP of 334.7 N ml CH₄/g VS (Set 1) and 362.9 N ml CH₄/g VS (Set 2), confirming its greater methane generation capacity. Raw spent wash yielded BMP values of 258.9 N ml CH₄/g VS (Set 1) and 300.2 N ml CH₄/g VS (Set 2). Among the co-digestion ratios, PM: RSW (1:2) performed best with BMP values of 289.8 N ml CH₄/g VS (Set 1) and 323.3 N ml CH₄/g VS (Set 2), demonstrating improved process stability. Notably, 90% of the total BMP was achieved within the first seven days across all substrate combinations, indicating rapid biodegradation kinetics.

Furthermore, a theoretical study evaluated the agronomic value of the organic modifications derived from the process. The application of Biomethanated Spent Wash (BMSW) and Compost/Press mud was found to reduce chemical fertilizer consumption in sugarcane cultivation by 4.8–6.4% and 13.0–19.8%, respectively, highlighting the dual utility of these by-products as renewable energy sources and organic soil amendments.

The findings of this study affirm that co-digestion of press mud and raw spent wash is a technically viable and environmentally sustainable approach for integrated waste

management, renewable energy generation, and reduction of dependence on chemical inputs in agriculture, thereby contributing meaningfully to the principles of the circular economy.

Chapter 1
INTRODUCTION

1. INTRODUCTION

1.1 General Background

The world's commodity markets make considerable use of coal, natural gas, petroleum and crude oil as energy sources, fuels, and chemicals. However, because fossil fuels take millions of years to develop on a global scale, their sources are finite and susceptible to depletion when exhausted. Research into the generation of alternative fuels from bio-resources has been sparked by the continuous consumption of fossil fuels and the impact of greenhouse gases on the ecosystem. In this context, biomass energy has emerged as one of the most promising prospects for the future of renewable energy sources. Globally, biomass is considered one of the largest renewable energy resources, with the potential to significantly contribute to the energy mix alongside solar and wind power. A renewable and green energy source is biogas. The energy used to produce biogas is obtained from the Sun through photosynthesis by plants, much like all other renewable energy sources. It is mostly composed of carbon dioxide (CO_2) and methane (CH_4), which can be purified to produce feed gas, liquefied to produce methanol, compressed to make vehicle fuel, and purified to produce methane for its use in producing heat or power. Anaerobic digestion of a range of raw materials, including agricultural waste, sewage sludge, energy crops, animal manure and food waste, can yield biogas. Methane-containing biogas is an important product that can be used to generate renewable sources of energy. Beyond energy production, the use of biogas offers crucial environmental advantages; it helps mitigate methane emissions from landfills and open-air manure storage (which is a much more potent greenhouse gas than CO_2) and it converts pathogen-containing organic waste into a valuable, odourless bio-fertiliser (digestate). This dual benefit of waste management and resource recovery enhances the overall economic viability of the anaerobic digestion process, making it a key technology in the circular economy model. Biogas can be used to generate heat and electricity. Furthermore, cooking, heating, lighting, and process heat can also be burned directly.

The biodegradation of organic material in an anaerobic decomposition process (without air) yields biogas. The anaerobic bacteria break down biological material and produce a variety of gases in different quantities, including CH_4 , CO_2 , and H_2S . 50% to 70% methane (CH_4), 30% to 40% carbon dioxide (CO_2), and trace amounts of other gases

make up the biogas combination. Biogas has a flame temperature range of 650°C to 750°C and is over 20 times lighter than air. This colorless gas is typically the same as LPG gas, which produces a crystal blue flame. The most commonly used and easily accessible feedstock source in India is cow dung. About 10 to 12 kilograms of manure can be produced daily by a single cow, and 20 to 22 kg can be produced daily by a buffalo. Since biogas is essential to its energy consumption and the slurry used as fertilizer in farms like compost (soil nutrients), there is no waste in it. Additionally, biogas can convert the gas's energy content into a variety of mechanical and thermal energy forms that can be used for diesel engine operation, heating, and electricity generation (Kukde Committee Report, 2012). These are the business and residential applications of biogas. In a chain of microorganisms known as methanogens, where methane generates bacteria that break down organic material and release stable products back into the environment, natural biogas creation is crucial to a biogeochemical carbohydrate cycle. Biogas and other sustainable renewable energy sources are the result of this process. Globally, microbial activity releases between 590 and 880 million tons of methane into the atmosphere annually. About 90% of methane is produced by biomass degradation, the remainder is produced by fossil fuel sources, such as petrochemical operations. Methane concentrations in the northern hemisphere are currently around 1.65 parts per million. The design, building, and operation of biogas facilities depend on an understanding of the fundamental mechanisms behind methane fermentation. The three bacterial processes that contribute to the anaerobic degradation process are hydrolysis, acidogenesis, and methanogenesis. The production of biogas is influenced by several factors, including variations in the temperature of the atmosphere, digester temperature, pH, and solid concentration. Bacterial activity is negatively impacted by each of these factors. India primarily pushes biogas technology for energy purposes. The most important issue is how to give energy to the vast population [9]. Advanced biogas facilities that are simple to operate have been established in India in addition to China. Biogas plants have been widely distributed to rural families during this time. With financial assistance, the National Biogas Development Project (NPBD) reached its 65th year in 1981 and achieved its objective of distributing household biogas plants in large quantities. However, it was greeted with a number of failures, as many plants were either underutilized or left idle. Numerous factors contributed to the biogas plant's failure, including the lack of established technological features, a limited distribution strategy, and an ignorance of the importance of customer education and

follow-up services. Nonetheless, the Indian government has persisted in encouraging the production of biogas. The production of biogas is directly promoted by the Ministry of Non-Conventional Energy Sources (MNES, formerly DNES). In 1982, the newly established DNES section assumed central authority over the delivery of biogas as a section of the Ministry of Power and Non-Conventional Energy Sources. Nearly a million home biogas plants began to operate in India in the interim, with 70–80% of those plants believed to be operational. The supply of biogas is directly promoted by the Ministry of Non-conventional Energy Sources (MNES, formerly DNES). One crucial issue was funding for the development of biogas. The Commissions concentrated on the conditions of the current biogas plants and also allocated the Department of Research and Development. The initiative also involves collaborations from several organisations. Nearly all states are in charge of the laws and policies that apply in their areas, which fall under the MNES-specified areas of assistance. Individual needs vary by area and have a significant impact.

1.2 Advantages of Biogas Technologies

Both the farmers engaged and society at large can gain economically and environmentally from the creation and application of the AD process for biogas generation. By strengthening the internal value chain of biogas production, local financial capability and employment protection in agricultural fields boost financial capability. Additionally, it raises living standards and promotes social and economic advancement. Biogas technologies provide substantial benefits across several sectors, most notably through environmental sustainability, economic growth, and agricultural enhancement. Environmentally, they serve as a crucial tool for climate change mitigation by capturing potent greenhouse gas (GHG) emissions.

1.2(A) Merits for the Society

A. Improved Public Health: Prolonged exposure to smoke and particle matter from conventional fuels can result in severe eye disorders as well as irritation and inflammation. This damage is avoided by using biogas to cook without smoke.

Severe burns are often caused by open fires and unsteady kerosene stoves, especially in women and children. Because they are robust and use a regulated

flame, biogas stoves greatly lower the risk of unintentional burns. Clean-burning biogas significantly lowers interior air pollution when it is used in place of conventional cooking fuels like firewood, charcoal, or kerosene. This has significant health benefits, especially for women and children who have respiratory conditions brought on by domestic smoke.

- B. Enhanced Sanitation and Hygiene:** Animal manure and sewage sludge are examples of organic waste that biogas systems employ. By efficiently eliminating germs and parasites, the anaerobic digestion process lowers the risk of waterborne illnesses and promotes better waste management.
- C. Odor Reduction:** The unregulated release of gases such as ammonia and hydrogen sulphide is the primary cause of unpleasant aromas from decomposing manure. By capturing methane and preventing its open emission, the sealed, anaerobic atmosphere of a biogas digester transforms it into odourless bio-slurry and clean fuel. The elimination of strong Odors from extensive waste storage enhances the overall perception and interaction between the farm or facility and its neighbours, making the environment more liveable for communities or farms close to heavily populated areas. Treating manure and organic waste in a sealed digester significantly reduces unpleasant Odors compared to open storage (like manure lagoons), improving the living environment for nearby communities.
- D. Energy Resilience:** Communities can increase local energy security and stability by reducing their need on centralized or imported energy sources by using their waste resources to produce electricity.
- E. Reduced greenhouse gas emissions and mitigation of global warming:** Carbon dioxide (CO₂) is released into the atmosphere by the burning of fossil fuels such as lignite, hard coal, crude oil, and natural gas, which have been stored in the Earth's crust for millions of years. Since carbon dioxide is a greenhouse gas, higher atmospheric CO₂ levels contribute to global warming. Combustion of biogas also releases carbon dioxide. However, the primary distinction is that, unlike fossil fuels, carbon from biogas has been extracted from the atmosphere through plant photosynthesis. As a result, the biogas carbon cycle is shut off in a fairly short period of time (between one and several years). By processing and using untreated livestock waste as fertiliser, AD's biogas generation further reduces the production of methane (CH₄) and nitrous oxide (N₂O).

- F. Waste reduction:** One of the primary benefits of using biogas as a substrate in AD is its ability to convert waste into a useful resource. The overproduction of organic waste from homes, farms, and factories is a major issue for the majority of European nations. An efficient method of implementing increasingly lenient national and international laws in this area is the development of anaerobic digestion, which uses organic wastes to produce energy before the digested substrate is composted as manure. Additionally, AD can lower the size of waste and the expense of disposing of it.
- G. Job creation:** The processing, distribution, and selection of AD feedstock, as well as the design, operation, and maintenance of biogas facilities, all demand work energy when using ADs for biogas production. Stated differently, the development of a domestic biogas industry boosts rural income, generates new jobs, and gives rise to new enterprises, some of which have significant economic value.

1.2(B) Benefits for the farmers

On July 4, 2023, Union Minister Nitin Gadkari stated that the government thinks farmers should be acknowledged as both "urjadata," or energy producers, and "annadata," or food producers. Gadkari claims that the current import bill is Rs 16 lakh crores. Farmers may receive this money straight into their homes if they grow crops that produce energy and can be turned into ethanol. Gadkari announced at a rally in Pratapgarh, Rajasthan, that "all vehicles will now run on ethanol produced by farmers." He asserts that if an average of 60% ethanol and 40% electricity were used, gasoline would be readily available for just Rs. 15 per liter, greatly benefiting the general populace. The following are some benefits of biogas technologies for farmers:

- 1. Flexibility to use different feedstock:** The capacity to use a variety of feedstocks, a concept known as co-digestion synergy, is the most important operational advantage of Anaerobic Digestion (AD), which improves system stability and resource yield for farmers. Although the AD process is sensitive to abrupt changes in pH or nutrient loading, the principal wastes, Press Mud (high solid content, slow to decay) and Raw Spent Wash (high moisture, frequently acidic), have complementary chemical qualities that are improved through

mixing. In order to maintain the optimal (C/N) ratio (usually 20:1 to 30:1) and alkalinity needed for steady methanogenesis, feedstock flexibility enables the system to use auxiliary substrates, such as locally accessible sugarcane waste, animal manure, or even energy crops. By reducing the possibility of fertilizer supply disruption brought on by seasonal fluctuations or operational imbalances at the sugar mill, this capacity guarantees the farmer consistent and dependable production of high-quality digestate. Additionally, by selling these balancing feedstocks to the AD plant, this flexibility turns the farmer's excess organic materials, which were previously disposal liabilities into tradable commodities, opening up new revenue streams.

- 2. Reduction of Odours and Pest Attraction:** Strong and disagreeable smells that are associated with the unregulated storage of industrial organic wastes are significantly reduced when anaerobic digestion is used, which is an important environmental and social benefit. High amounts of volatile solids (VS) and sulphur compounds are found in raw spent wash and press mud. When these materials decompose openly, they emit harmful gases including ammonia and hydrogen sulphide, which are the main causes of foul Odors. By keeping the organic material inside a sealed reactor, turning the volatile solids into methane (biogas), and producing an output digestate (slurry) that is nearly odourless and unappealing to pests, AD successfully internalizes this pollution. By simultaneously disrupting the reproduction cycles of flies and other insect vectors, this confinement and stabilization procedure can reduce odours by up to 80% or more, enhancing the environment and public health in the vicinity of the sugar mill and other farmlands.
- 3. Additional Income for Farmers and Enhanced Social Role:** By digesting Press Mud and Spent Wash, Anaerobic Digestion (AD) is integrated into the sugar business, resulting in a significant financial change that generates additional income for the nearby farming community. The main sources of this income are revenue diversification and cost displacement, which involves replacing pricey mineral fertilizers with locally obtained digestate. Farmers gain chances to monetise their own agricultural leftovers (such as animal manure or cane waste) by sending these feedstocks to the AD plant for steady co-digestion, and they also directly benefit from lessening their reliance on erratic external input markets. As a result, the farmer takes on a new and crucial social role that

goes beyond traditional food production: they become important players in the local circular economy, acting as dependable feedstock suppliers and vital nutrient cyclers that guarantee the green energy project's financial and environmental sustainability.

- 4. Enhanced Veterinary and Environmental Safety:** The significant improvement of veterinary and environmental safety on agricultural sites that apply the resulting digestate is a crucial, but frequently disregarded, advantage of using biogas technology. The AD method exposes the feedstock, Press Mud and Spent Wash to a controlled sanitation process, in contrast to the direct land application of untreated industrial or agricultural wastes, which entails the risk of pathogen and disease transmission. This controlled treatment is intended to inactivate or eliminate pathogens, bacteria, weed seeds, and other biological risks. It is accomplished either by means of particular thermophilic temperatures or a minimal assurance of the storage period within the digester. As a result, the farmer receives a stabilised digestate that is clearly safer and better suited for use as a biofertilizer than raw materials, which lowers the risk of disease transmission to animals and lessens the veterinary hazards related to nutrient cycling.

1.3 Anaerobic Digestion Process

AD is a naturally occurring process that produces a gas mixture known as biogas from raw materials like industrial and municipal wastewater, agro-industrial, municipal, food activities, and vegetable wastes. It typically takes place in watercourses, sediments, and wet soils in the absence of oxygen [1][2]. Currently, AD is the only biological treatment method that can generate biogas, which is a mixture of carbon dioxide (CO_2) and methane (CH_4). In place of composting, AD permits organic materials to biodegrade in the absence of oxygen, producing a digested product suitable for use as soil fertiliser. Usually, the AD process takes place in a closed vessel. The biogas that is produced is sent to interim storage before being used in final applications. Heat and electricity production are biogas's primary commercial uses. Digestate, or leftover solids and organic materials, will be present in the vessel following AD. It is possible to divide digestate into liquid and solid streams. Both streams can be used as fertilizer in agricultural settings because they contain important plant nutrients[3]. The AD process is a dynamic, intricate system

that involves physical-chemical, biochemical, and microbiological activities. AD has been found to be the most environmentally friendly technique of treating biowaste because it allows for the production of bioenergy and potential byproducts like soil biofertilizer in addition to keeping biodegradable materials out of landfills[4]. The AD process involves a number of phases and intricate interactions between several microbes. At various phases, diverse microbial populations work together to break down the complex biomass polymers and transform them into a gaseous mixture. At particular stages, some bacterial groupings are more active than others. One stage's output serves as the input for subsequent stages.

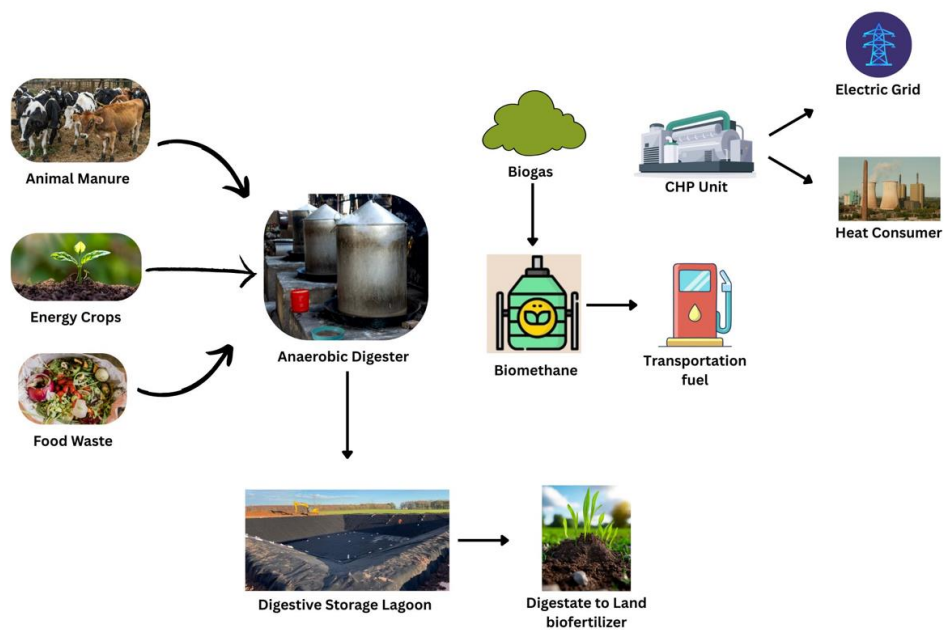
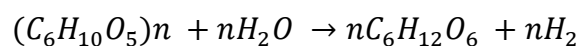


Figure 1.3: AD PROCESS

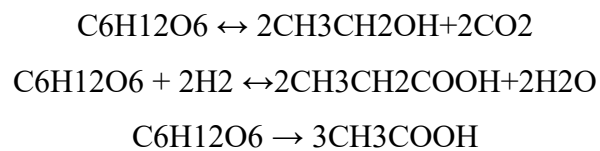
The biological AD process can be separated into four independent stages: i) hydrolysis, ii) acidogenesis, iii) acetogenesis, and iv) methanogenesis.

1) HYDROLYSIS: In essence, this is the initial phase of digestion. The complex polymeric structure of cellulose, starch, and proteins is broken down by water and extracellular enzymes into their corresponding simple units (monomers or oligomers), such as glucose, fatty acids, and amino acids. Equation (1) illustrates the general hydrolysis reaction,

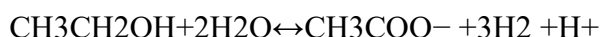


Amylase, cellulase, lipase, protease, and pectinase are examples of hydrolytic enzymes. Hydrolytic bacteria typically develop extremely quickly. However, the breakdown of polymers enters a rate-limiting step for substrates rich in lignin. While most chemicals at this stage require additional breakdown through other stages, some are ready to be turned into biogas.

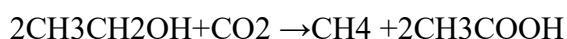
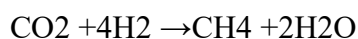
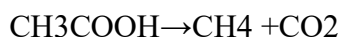
- 2) ACIDOGENESIS:** Acidogenic bacteria further break down the hydrolysis products in the acidogenesis stage. Alcohol (ethanol, methanol), ketones (glycerol and acetone), and short-chain volatile fatty acids (VFA) (acetic acid, propionic acid, formic acid, and lactic acid) are the major results of hydrolysis. Alcohols, CO₂, H₂, NH₃, and trace amounts of other chemicals are also produced as byproducts. At the last stage, the methanogens can easily use certain products like CO₂, H₂, acetate, and formate. Acidogenesis is generally a very fast process, and there is a risk of VFA accumulation in the digester, resulting in digester toxicity if not properly controlled. Below equations illustrate the chemical reactions in the acidogenesis stage.



- 3) ACETOGENESIS:** Acetogenic bacteria produce acetate, CO₂, and H₂ from the byproducts of the acidogenesis stage and some long-chain fatty acids from the hydrolysis step. If the partial pressure of H₂ is more than 10⁻⁴ atm, reactions in the acetogenesis stage are not thermodynamically spontaneous. However, by eating the generated H₂, methanogenic bacteria reduce this partial pressure. The acetogenesis stage is thermodynamically possible due to this syntrophic relationship, in which some bacteria are fed from the products of other bacteria. This interspecies H₂ transfer is synonymous with electron transfer as H₂ is essentially a proton (H⁺) with an additional electron. The rate of this electron transfer can significantly influence the overall digestion rate. Below equations illustrate the chemical reactions in the acetogenesis stage.



- 4) METHANOGENESIS:** All of the intermediate products from the other stages are converted into methane in this last stage. Because methanogenic bacteria cannot thrive in the presence of oxygen, this stage is strictly anaerobic. Two distinct types of bacteria, acetophilic and hydrogenophilic, convert CH_3COOH (acetate) and H_2 into CO_2 and CH_4 . Hydrogenophilic bacteria transform H_2 and CO_2 into CH_4 , whereas acetophilic bacteria transform acetate into CH_4 and CO_2 . The reactions in this stage is illustrated below,



1.4 Lignocellulosic biomass

The environment is suffering from climate change, which is made worse by resource overuse, which raises greenhouse gas emissions worldwide[5]. To fulfil the world's future energy needs, sustainable and ecologically friendly energy derived from renewable resources is needed. In order to manufacture second-generation biofuels and other biobased chemicals without jeopardizing global food security, lignocellulosic biomass (LB) continues to draw attention from throughout the world as a sustainable substitute for fossil carbon resources[6]. Among them are agricultural wastes like bagasse and cereal straw[7], forest leftovers like pine, specialized crops, and short-rotation coppices like miscanthus, among others[8]. lignocellulosic biomass (LCB) is a plentiful feedstock that is regarded as an exceptionally large resource for the production of chemicals and biofuels with global yield of 1.3billion tons per annum [9]. First-generation substrates utilized to generate biofuel included wheat and sugarcane, which are used to make food goods. Overuse of food crops as feedstock for the production of biofuel results in increased land use and ecosystem losses[10]. Therefore, after significant research, second-generation biofuels like LCB residues were recognized. The production of biofuel from lignocellulosic biomass (LCB) is a promising method with lower carbon

emissions[11]. The LCB contains the leftovers from the paper, forestry, and agricultural industries[12]. There are numerous benefits to converting trash into energy or fuel, including lower greenhouse gas emissions, waste reduction, revenue generation, and waste processing[13]. Cellulose, hemicelluloses, and lignin make up the majority of LB, which is a complex polymer assembly that is inherently resistant to enzymatic conversion. Because of this, certain pre-treatment procedures are required to increase cellulose accessibility by altering the chemical and/or physical structure of LB and promoting the transformation of polysaccharides into fermentable sugars[14].

1) CELLULOSE: It is the main component of plants that is utilised as a valuable source for storing vast amounts of energy through photosynthesis. These linear polymers, which are present in hardwood parts, have the characteristics of being hard and water insoluble. It mostly consists of 3000–14000 glucose monomers connected by a 1, 4 glycosidic linkage[15]. The cellulose contains roughly 60–70 polymers that are joined by hydrogen bonds to create the cellulosic structure of fibre crystals[16]. The main cause of the limitation in biomass hydrolysis is these hydrogen bonds. The degree of crystallinity rises with hydrogen bond orientation, which ultimately causes problems with cellulose biodegradation and is entirely dependent on the kind of plant[17]. It functions as a versatile source for the production of biofuels and specific chemicals since it is insoluble in water, has a higher tensile strength, is biocompatible, hydrophilic, and has a reactive hydroxyl group[18]. This component's carelessness increases the cellulose's surface area, which prevents cellulose breakdown. It is difficult to break down biologically because the inadequately hydrated region limits the activity of hydrolysis enzymes[19].

2) LIGNIN: The main obstacle to LCB accessibility in the case of biofuel production is lignin, a heteropolymer that covalently joins cellulose and hemicellulose in a three-dimensional structure and creates a protective covering. It is an aromatic polymer that adds significant structural support, impermeability, and resistance to hydrolysis and degradation to plant biomass[20]. Peroxide-interceded dehydrogenation of the phenyl propionic alcohols during polymer synthesis produces free radicals. The lignin must break

down via co-metabolism in order to get to the cellulose and hemicellulose components[19]. Gaiacyl, p-hydroxyphenyl, and syringyl are the three main phenolic chemicals that make up lignin. Pretreatment is required for the polysaccharide recalcitrance behaviour in plant biomass with a higher lignin concentration in order to maximize fuel and energy production[15].

- 3) HEMICELLULOSE:** The second main component of the heterogeneous polymer hemicellulose is a variety of polysaccharides, including pentose, xylan, glucomannan, hexose, and galactomannan. It is connected by the glycosidic bond (1–4)[21]. The hydrogen connection that connects cellulose, lignin, and hemicellulose gives the cell wall structure flexibility. The hemicellulose structure was broken down by the biologically generated enzymes[19]. Based on their properties, such as a lower degree of polymerization and no crystallinity, the polysaccharide hemicelluloses were broken down into monosaccharides[22]. Methane generation is possible when hemicellulose breaks down into sugar and acetic acid[23].

1.5 Anaerobic Digestion Feedstock

Anaerobic digestion was primarily designed to treat liquid waste, such as sludges from organic or physico-chemical processes, manure, and wastewater from homes or businesses, with or without suspended particulates. The need for anaerobic digestion has grown in recent years due to increased environmental consciousness as well as the need for renewable energy and sustainable waste management practices. The main purpose of anaerobic digestion was to handle liquid waste, including manure, sludges from organic or physico-chemical processes, and wastewater from houses or companies, with or without suspended particles. The demand for anaerobic digestion has surged recently as a result of growing environmental awareness, the need for sustainable waste management techniques, and the requirement for renewable energy.

1.5(A) Different Bio-methanation Substrates and Co-Substrates

Organic resources are essential as a starting point for the production process and as feedstock to produce biogas. Biogas can be produced using a variety of feedstocks, including:

- a) Cow dung
- b) Municipal Solid Waste
- c) Food & Vegetable Waste
- d) Poultry Waste
- e) Agricultural Waste
- f) Piggery Waste
- g) Paper Waste
- h) Sewage Sludge
- i) Energy Crop

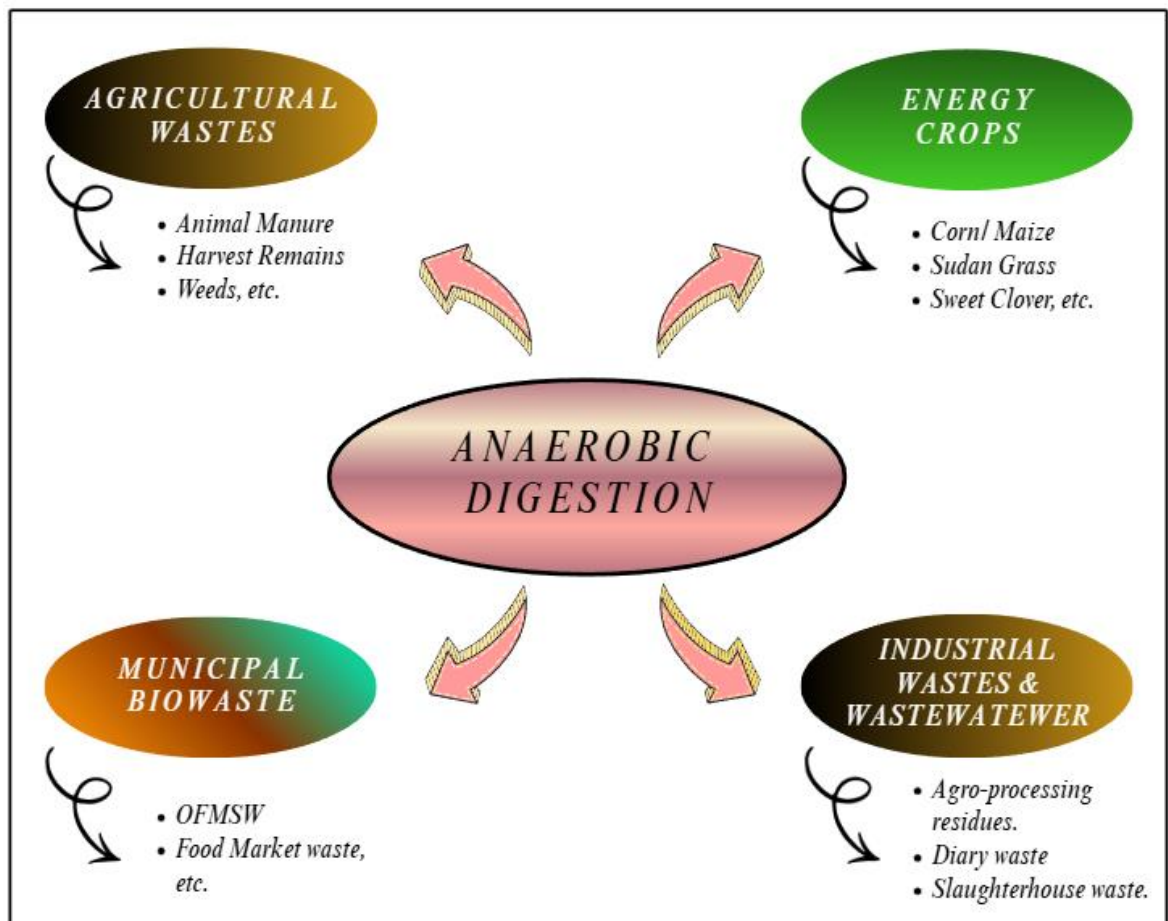


Figure 1.5 (A): Feedstocks for Anaerobic Digestion

- a. **Agricultural Wastes:** Animal dung, harvest remnants (such as crop stalks and straw), and weeds, all easily obtained from farming operations, fall under this category.

A typical fundamental feedstock is animal dung, which provides a steady supply, essential trace nutrients, and superior buffering ability for a stable AD process.

- b. **Energy Crops (e.g., Corn/Maize, Grass):** Sweet clover, Sudan grass, and corn/maize silage are examples of energy crops that are grown especially to maximize biomass and feed into the AD process. They offer a reliable, dependable, and high-quality supply of feedstock with known methane outputs. These feedstocks have a great potential for producing biogas since they usually include high levels of readily degradable carbohydrates.
- c. **Municipal Biowaste (e.g., OFMSW, Food Market Waste):** This comprises domestic food waste, food market waste, and the organic fraction of municipal solid waste (OFMSW). It has one of the greatest methane outputs per unit mass and is distinguished by a high organic content.
- d. **Industrial Wastes and Wastewater:** High-strength liquid and solid wastes from businesses including slaughterhouses, breweries, dairies, and agro-processing, are included in this broad group. These wastes have a large potential for energy recovery since they are frequently highly concentrated (high chemical oxygen demand (COD)).

1.6 PRESSMUD

One of the most popular raw sources for making sugar and ethanol is sugar cane. Bagasse, press mud, spent wash, and cane residue left in the field after cane harvesting are the four main categories of waste products produced by the sugar processing and alcohol making industries[24]. The leftover material from the cane crushing process is called bagasse, and it serves as the industry's fuel[25]. Press mud is a waste product of the compressed sugar industry that is created after cane juice is filtered. It is an excellent source of fertiliser. Approximately 12 million tons of press mud (Filter cake) are produced by Indian sugar mills as a byproduct of double sulphitation procedures. Press mud is typically sold to farmers as immature compost or dumped on open fields. Because dry press mud has a higher concentration of combustibles, it can be used to generate energy. Because it contains roughly 5–15% sugar, it is also a good producer of biogas[26].

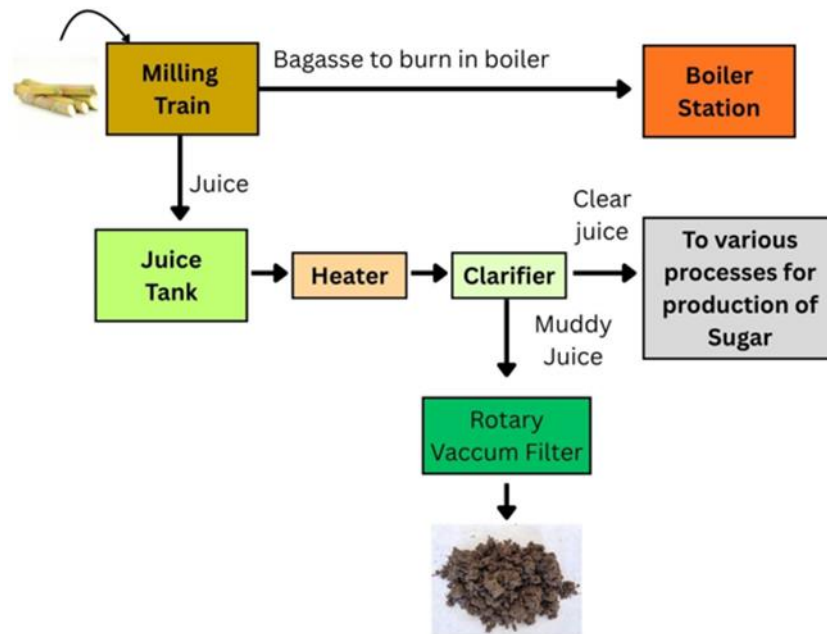


Figure 1.6: A schematic diagram of the formation of press mud waste in a sugar mill.

Press mud is a solid residue that is left over when sugarcane juice crystallizes. In India, press mud is typically utilized as manure. It is an amorphous, lightweight, spongy, and dark brown to black substance. It typically has 60–85% moisture (w/w), the chemical makeup varies according on the type of cane, the state of the land, the nutrients used in the field, the clarity method used, and other environmental conditions. Press mud from sugar factory typically contains 71% moisture, 9% ash and 20% volatile solids, with 74-75% organic matter on solids.



Figure 1.6.: Press Mud

1.6.1 Press Mud Composition (Based on Dry Solids)

Press mud's composition reveals the particular organic fractions that determine its breakdown kinetics during anaerobic digestion (AD) and contribute to its energy potential. The substance has a complicated structure and contains a substantial amount of lignocellulosic material, including cellulose (11.4%), hemicellulose (10.0%), and lignin (9.3%). Longer hydraulic retention durations are required for full gas recovery since these components are resistant to quick microbial attack. On the other hand, the press mud is abundant in easily accessible minerals and high-energy fractions, such as residual fermentable sugar (5.7%) and protein (15.5%). This high concentration of quickly degradable components suggests a high potential for initial biogas generation, but it also raises the possibility of process instability because of the buildup of volatile fatty acids (VFAs). Wax (7.2%) is a noteworthy non-lignin organic fraction that greatly increases the heating value but needs to be carefully managed in the digester to avoid operational problems. Lastly, the low sodium (Na) level (0.22%) indicates a low likelihood of salinity-induced inhibition[27].

1.6.2 Press Mud Characteristics (Physical and Chemical)

Press mud's handling qualities and suitability for anaerobic processing are determined by its general features. With a high moisture content of 76.3%, it is essentially a wet material. To obtain the ideal solid concentration for steady digester operation, this high-water content requires either dewatering or careful dilution planning. A high volatile matter content of 76.6% (on a dry basis), which is closely correlated with the amount of material convertible to biogas, demonstrates the superior quality of the material for bioenergy. Chemically speaking, its high potential energy density is confirmed by the presence of sugars (6.4%) and wax (7.2%). The C/N ratio is at a low value of 14, which is critical for digester performance. Methanogenesis typically operates best between 20:1 and 30:1; this ratio is not ideal. This feature is the main reason why co-digestion with carbon-rich feedstocks is necessary to guarantee process stability and optimise methane production[27].

1.6.3 Advantages of Press Mud

- a) High Biogas Yield Potential:** When used in Anaerobic Digestion (AD), press mud's abundance of easily accessible volatile solids—mostly leftover sugars and proteins—ensures a high potential for the production of methane-rich biogas.

- b) Excellent Source of Organic Carbon:** It is a useful material for creating the humus-like residue required to improve soil fertility and structure because of its high organic matter content (typically 70% volatile solids).
- c) Abundant and Consistent Supply:** Press mud is a major byproduct of the sugar industry and is produced in high, consistent amounts (3–4% of all cane crushed) associated with seasonal crushing, guaranteeing a steady supply of feedstock.
- d) Nutrient-Rich Fertilizer:** Its digested or composted form is a better organic substitute for manufactured chemical fertilizers since it contains vital macronutrients including nitrogen, phosphorus, and potassium.
- e) Improves Soil Physical Properties:** Press mud application greatly improves soil properties, such as improved soil structure, higher aeration, and better water-holding capacity.
- f) Effective Waste Management Tool:** An environmentally sound way to manage a sizable industrial waste stream is to use press mud through AD or composting, which avoids landfilling and related problems like leachate and Odors.
- g) Economical Co-substrate:** It is a cheap feedstock that, by utilizing its buffering capability, can be effectively co-digested with other wastes (such as carbon-rich crop residues) to increase the overall stability and efficiency of the digester.
- h) Source of Valuable Wax Recovery:** Sugarcane wax, which can be removed either before or during processing, makes up a significant portion of press mud. This wax finds use in high-value industrial applications such as polishing and cosmetics.
- i) pH Neutralization and Buffering:** Press mud's natural properties can provide a more stable microbial habitat by buffering pH variations in acidic soils or AD systems.
- j) Resource Circularity and Sustainability:** Press mud use fully reflects the ideas of a circular bioeconomy, minimizing environmental effect and producing value products by turning this industrial residue into renewable energy and soil nutrients.

1.7 RAW SPENT WASH (VINASSE)

The sugarcane industry produces a liquid waste known as Distillery Spent Wash (DSW). It has a lot of inorganic and organic materials in it. Additionally, DSW has adequate levels of micronutrients (zinc, copper, iron, and manganese) and macronutrients

(nitrogen, phosphorus, potassium, nitrogen (N), phosphorus (P), potassium (K), calcium (Ca), magnesium (Mg), and sulphur (S), all of which enhance crop growth and yield.[28].



Figure 1.7: Raw Spent Wash (Vinasse)

A by-product of the sugar extraction process, molasses is the most widely utilised raw material in distilleries for the production of bioethanol[29]. Following alcohol distillation, a significant amount of highly coloured spent wash is produced, which has a high organic load (80,000–100,000 mg/l for chemical oxygen demand and 40,000–50,000 mg/l for biochemical oxygen demand), is dark brown in colour, has a low pH, and is heated to a high temperature[29]. A variety of organic substances, including polysaccharides, reduced sugars, lignin, proteins, melanoidin, waxes, etc., are responsible for the distillery spent wash's high chemical oxygen demand (COD) and BOD [29].

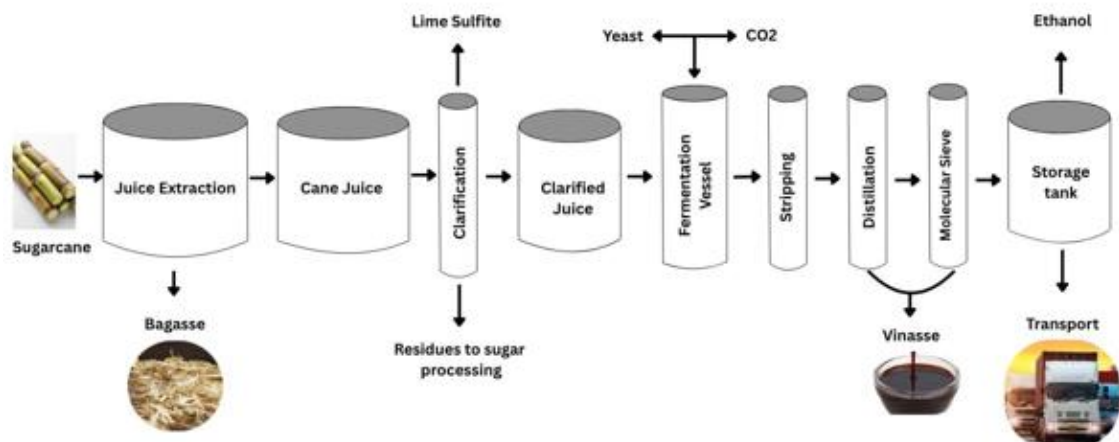


Figure 1.7: The process of producing vinasse from sugarcane.

1.7.1 Raw Spent Wash Characteristics

Colour: Dark brown is the description given to each sample. Because of the high levels of melanoidin colours and caramelized sugars produced during the fermentation of molasses, this is common for distillery wasted wash [29].

Total Dissolved Solids (TDS): The observed range is 12,600 mg/l to 112,400 mg/l. large levels of dissolved organic and inorganic stuff, mostly from salts, molasses, and nutrients, are indicated by high TDS. Because high TDS might hinder biological activities, treatment becomes difficult [30].

Total Suspended Solids (TSS): ranges between 2,000 and 40,000 mg/L. Cell debris and unfermented organic waste make up the majority of suspended solids. Sedimentation or screening may be necessary prior to treatment since high TSS can clog treatment systems [31].

pH: It often falls from 2.8 and 5.5, signifying an acidic environment. Organic acids produced during fermentation and leftover molasses are the cause of low pH. Neutralization is frequently required before biological treatment of acidic wastewater [32].

Chemical Oxygen Demand (COD): It ranges from 8,000 mg/L to 1,580,000 mg/L which is really high. COD calculates how much oxygen is needed to chemically oxidize both organic and inorganic substances. Large concentrations of both biodegradable and non-biodegradable organics are indicated by high COD [28].

Biochemical Oxygen Demand (BOD): ranging from 35,000 to 96,000 mg/L (sometimes greater in references). shows a high level of organic material that decomposes. An indication of biodegradability is provided by the BOD to COD ratio: $BOD/COD \geq 0.5$ indicates high biodegradability. The wastewater is partially biodegradable but still contains refractory chemicals, as indicated by the BOD/COD values, which are often between 0.4 and 0.6. [33][34].

1.8 Anaerobic Co-Digestion

The breakdown of two or more organic substrates is known as anaerobic co-digestion (AcoD), which gives the anaerobic fermentation process a synergistic impact that increases the production of biogas[35]. This method shows promise for increasing the outputs of methane and biogas from the anaerobic digestion of solid wastes[36]. The AcoD method is beneficial because of its increased methane yields, economic feasibility, and ability to resolve some of the problems that occur during mono-digestion. AcoD is a prominent area of study for enhancing conventional AD technology because of these problems, which include imbalanced nutrients and the presence of inhibitors and resistant chemicals in the substrate[37]. The substrate's C/N ratio is one factor that has a big impact on how the AD process develops. The ideal substrate ratio is 25:1; higher C/N ratios lead to lower methane concentrations in biogas, whereas substrates with a C/N ratio that is too low can cause ammonia to build up in the digester and prevent the synthesis of methane[38]. Co-digesting carbon-rich substrates, such as crop wastes, with nitrogen-rich substrates, like animal manure, can produce an appropriate C/N ratio in the anaerobic digester[39]. It has been demonstrated that AcoD offers a more compelling financial argument for the installation of CHP (combined heat and power) systems, which are thought to be the most cost-effective way to produce energy from biogas[40]. By increasing the nutrients available to microorganisms and the organic loading rate, AcoD enhances substrate digestion and energy production while lowering the toxicity of chemical inhibitors through co-substrate dilution[41].

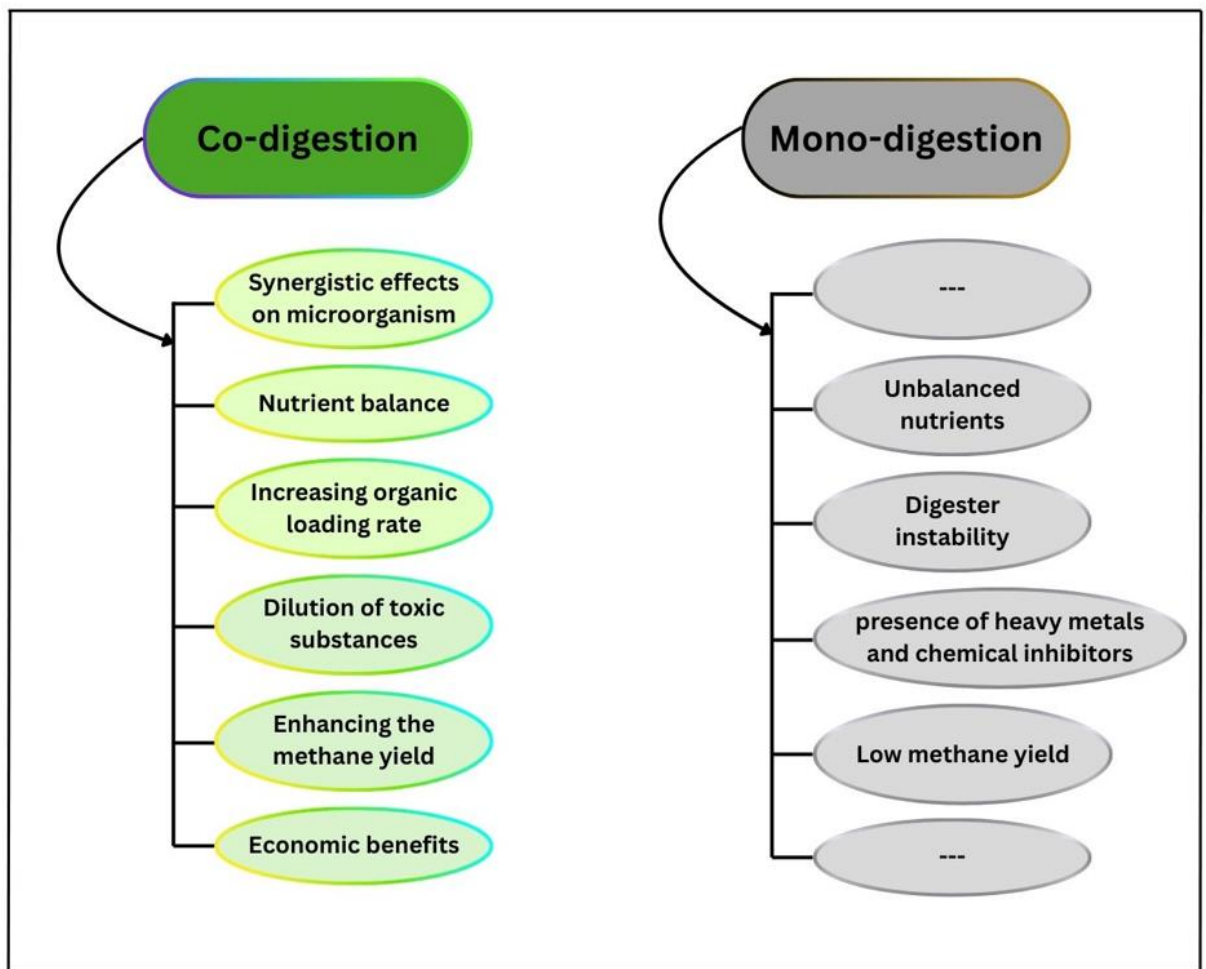


Figure 1.8. Advantages of Co-digestion technology compared with mono-digestion process[42]

1.9 Biochemical Methane Potential (BMP)

The greatest amount of methane that can be produced by anaerobic digestion is determined by the biomethane potential, also known as the biochemical methane potential (BMP), of a particular substrate. The definition describes the sample's current condition as it is examined or used in fermentation procedures. Anaerobic digestion plant substrate characterisation and efficiency assessment depend heavily on the BMP. It allows for dependable process balance and is essential for evaluating the quality and financial worth of various substrate types. In a lab setting, anaerobic batch tests are commonly used to assess BMP (BMP test).

The basic idea of a BMP test is simple: An active and functional inoculum is used to digest a sample of the examined substrate in bottles that are incubated at the proper temperature. Furthermore, blank samples that solely contain the inoculum are also

incubated. Subtracting the methane production of the blank (inoculum) from the substrate sample (substrate + inoculum) yields the resulting methane production of the substrate, assuming no synergistic or antagonistic effects. The experimental BMP of the substrate is defined as the final value of cumulative methane production following test termination.

1.10 Problems and Difficulties in the Production and Use of Biogas.

The supply of biodegradable goods is limited to sites of deposition when anaerobic digestion from any feedstock is realized in rural regions as a step forward for sustainable waste disposal. Their viability is dependent on the industry's needs for their finished products (gas, manure, or compost) as well as the institutional and technological setup of each unit. Biogas generated from the disposal of locally supplied substrates appears to be a solution to the energy problem in the majority of isolated areas of developing countries. The needed fuel temperature is about 20% higher than the power produced instantly by burning the entire amount of dung; the fuel supply uses around 25% of the dung. The primary cause of this is the extremely high frequency of use (60%) in contrast to the poor efficiency (11%) of burned dung cakes. Thousands of biogas reactors were constructed in developing countries. Several basic, chemical, microbiological, manufacturing, and social concerns need to be resolved in order to ensure the widespread deployment of biogas programs from energy crops with guarantees of economic efficiency and public recognition. The construction of community-based biogas plants, efficient use of outlet manure, increased bacterial conversion of CO_2 to CH_4 , adoption of pre-treatment methods for the degradation of fibrous content in lignocellulosic biomass, advanced methods for converting urban waste into energy, optimal plant layout, efficient burners, solar-based digesters, and the connection of biogas technologies to other unconventional sources of energy are some of the important variables that indicate that effectiveness in service needs to be established. It was also discovered that households frequently construct large-scale reactors, even if they are solely utilised for cooking and not for broader energy needs. It was found that a large plant produced enough substrates and ultimately failed to produce gas. It has also been demonstrated that under-collection of the dung, which was generally thought to be between 30% and 40% of the necessary capacity and was primarily caused by cattle working in the field, occurs during feeding, which would lower gas output. Dung accessibility may also vary. As mentioned earlier, elements of

climatic disturbance include droughts that can reduce the supply of dung, forced livestock sales, or even the destruction of bovine animals.

1.11 Need for Research

- a) Despite being generated at the same location, press mud and raw spent wash are usually treated separately in independent systems, resulting in limited information on their combined treatment performance.
- b) Since both substrates are available at a single location, they offer a clear logistical advantage for co-digestion.
- c) Although co-digestion is known to improve biodegradability and biogas generation compared to individual digestion, the optimum mixing ratio for these substrates has not yet been established.
- d) Write about Fertiliser reduction

Chapter 2

AIM AND OBJECTIVES

2. Aim and Objectives

1. To characterize the physicochemical properties of press mud and raw spent wash individually, including parameters such as pH, Total Solids (TS), Volatile Solids (VS), Chemical Oxygen Demand (COD), Biochemical Oxygen Demand (BOD), C/N ratio, and moisture content, to establish a baseline understanding of their digestion potential.
2. To assess the individual (mono-digestion) biochemical methane potential (BMP) of press mud and raw spent wash separately under controlled anaerobic batch conditions, to serve as a reference for evaluating the synergistic effect of co-digestion.
3. To investigate the effect of different co-digestion ratios (RSW only, PM only, 1:1, 1:2, and 1:3 of PM: RSW) on cumulative biogas and methane yield, in order to identify the optimal substrate mixing ratio for maximum energy recovery.
4. To evaluate process stability parameters, including pH variation, volatile fatty acid (VFA) accumulation, and alkalinity across all co-digestion ratios to determine the most stable operating condition for anaerobic digestion.
5. To analyse the methane content (%) in the produced biogas across different mixing ratios using gas chromatography or equivalent methods, and to determine which ratio yields the highest quality biomethane.
6. To recommend the most effective and sustainable co-digestion ratio for simultaneous maximization of biogas production, process stability, organic waste treatment, and digestate quality, contributing to the circular bioeconomy model for the sugarcane processing industry.
7. To characterize the nutrient composition of the digestate produced from co-digestion including macronutrients and micronutrients and to quantify its bio-fertilizer value as a direct substitute for chemical fertilizers, thereby assessing the potential reduction in synthetic fertilizer application in sugarcane farmlands.
8. To evaluate the potential of co-digestion, digestate in reducing chemical fertilizer dependency by comparing the nutrient availability (N, P, K) in the digestate with standard recommended doses of chemical fertilizers (such as Urea & DAP) used in sugarcane cultivation, and to estimate the proportion of synthetic fertilizer that can be replaced by digestate application without compromising soil health or crop yield.

Chapter 3
LITERATURE REVIEW

3. LITERATURE REVIEW

3.1 INTRODUCTION:

A natural biochemical process known as anaerobic digestion (AD) turns organic waste into biogas that can be burned. Historically, anaerobic digestion plants were first constructed in South Asia in the eighteenth century, and as technology advanced, they were gradually brought to Europe; however, it wasn't until 1930 that the true process of producing biogas from the breakdown of organic materials was fully understood. Fundamentally, AD is a sequence of metabolic events where bacteria break down the organic materials of any substrate into a gaseous mixture (CH_4 , CO_2 , H_2 , H_2S , etc) in the absence of free oxygen. This process is a dynamic, intricate system that includes physical-chemical, biochemical, and microbiological activities. All things considered, AD is a sequential procedure executed by several bacterial and methanogenic archaeal groups, involving four key steps in converting a substrate into distinct products: hydrolysis, acidogenesis, acetogenesis, and methanogenesis. The technology of AD can be classified according to the dry matter (DM) of the initial substrate as: wet digestion, where the substrate must have a DM content below 10%; or dry digestion, in which the substrate should have a DM content exceeding 20%. Although AD has long been used to manage waste in cities and agriculture, it is currently receiving increased interest as a potential alternative energy source. Specifically, by recovering methane-rich biogas, encouraging environmentally friendly waste management techniques, and lowering greenhouse gas (GHG) emissions, the anaerobic digestion process has proven to be a viable alternative biomass conversion pathway for diversifying the product portfolio of sugarcane biorefineries[3].

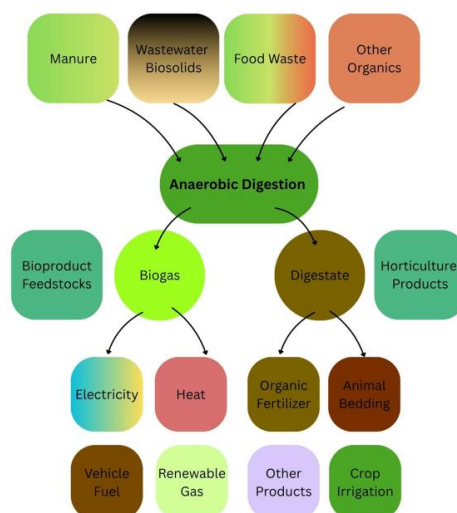


Figure 3.1. Flow of feedstocks through the AD system to produce biogas and digestate.

Press mud is a solid byproduct derived from sugarcane juice prior to the crystallization of sugar. In India, press mud is typically utilized as manure. It is a dark brown to black material that is amorphous, squishy, spongy, and lightweight. It typically has between 60 and 85 percent moisture (w/w), the chemical makeup varies according on the type of cane, the state of the soil, the fertilizer used in the field, the clarity method used, and other environmental variables [27]. It regulates soil pH, enhances drainage, stops soil erosion, crusting, and cracking, and encourages healthy bacterial and microbial development. Additionally, it serves as a soil conditioner and reclaimant. Due to its larger amount of combustibles, dried press mud can be used to generate energy[43]. Applying sugar filter cake as manure to the soil improves yield because it is high in nutrients. It is determined that the amount of organic carbon has increased. It also causes the amount of clay and moisture to rise. Under some circumstances, the addition of treated press mud to the soil results in an increase in other nutrients like phosphorus and nitrogen while causing a slight fall in potassium levels. Depending on the fertilizer content, the analysis reveals a comparable improvement in the soil's electrical conductivity, which lowers the uptake of iron by plants in particular conditions[44].

Due to the extensive industrial uses of alcohol in industries including food, medicine, fragrance, and more, the number of alcohol distilleries is rapidly increasing. It serves as an alternative fuel as well. The undesirable residual liquid waste produced during the

production of alcohol is known as distillery spent wash, and the pollution it causes is one of the most serious environmental problems. Untreated or inadequately treated sewage frequently makes its way into water courses despite effluent quality guidelines. Water quality in many parts of the world is seriously threatened by distillery effluent, which has a distinctively disagreeable smell. DSW is an undesirable, dark brown liquid with an offensive Odor that is produced by the distilleries reboiler. Approximately 12 to 15 liters of DSW are generated for every liter of alcohol on average. Consequently, 285 distilleries nationwide produce about 40 billion liters of wasted wash annually. DSW typically comprises low pH, inorganic salts, caramelization products, and recalcitrant components with high COD and BOD as well as chemical oxygen demand. The unscientific disposal of DSW in open drains and landfill in the vicinity of distilleries, creates several environmental issues due to its toxic nature[45]. The demand for novel technology to handle this effluent effectively and affordably has been sparked by the growing generation of distillery wasted wash on the one hand, and strict legal regulations regarding its disposal on the other. Numerous cleanup procedures have been implemented, and new bioremediation techniques are being developed to handle distillery waste wash.

Examining the current state of research on anaerobic digestion, biogas production, and waste product utilization, with a focus on press mud and distillery wasted wash is the goal of this literature review chapter. In order to evaluate primarily sustainable feedstock materials, it focuses on studying a broad variety of the readily available literature, gathering and analysing data on anaerobic processes. Additionally discussed are the biochemical processes, the potential and efficacy of various feedstocks, and the economic and environmental benefits of integrating waste products into biogas production systems. Through a comprehensive review of the literature, this chapter seeks to highlight the potential of press mud and distillery waste wash as a sustainable feedstock for the generation of renewable energy and to pinpoint areas in need of additional study and advancement. The focus of this literature review is on:

1. Anaerobic digestion
2. Substrates & Co-substrates
3. Analysis Parameters
4. Outcomes Analysis

3.2 Anaerobic Digestion

3.2(A) Historical Background

The study's historical records indicate that the anaerobic digestion (AD) process is among the oldest methods. In Assyria, biogas was used as a fuel to heat bathwater in the tenth century BC. Scientific study in the 17th century led to advancements in the AD. The evolution of combustible methane from organic matter was made possible by Jan Baptista Van Helmont. Additionally, Count Alessandro Volta discloses that a study was conducted in 1776 on the amount of combustible gas production and decomposing organic waste. Industrial AD uses began in 1859 with the establishment of the first anaerobic digestion facility in Bombay, India. Buswell and colleagues discovered anaerobic bacteria and environments that promote methane production in the 1930s. Anaerobic digestion was a popular technique for disposing of sewage since it produces more methane and has a high removal effectiveness. These days, China and India have a propensity toward larger, more sophisticated agricultural technology with better control over biogas energy production processes. Both domestic and commercial trash treatment are currently employing this method. Pollution caused by the use of traditional AD technologies has decreased in Taiwan due to the prompt disposal of waste from animal husbandry. Due to two significant factors, Europe has recently been influenced to research the AD process: i) high energy costs and ii) stringent environmental regulations, especially with regard to monitoring the disposal of organic matter into reserve areas.

3.2(B) Anaerobic Digestion Process

AD is a naturally occurring process that uses raw materials such as municipal and industrial wastewater, agro-industrial, food, and vegetable wastes to create a gas combination called biogas. In the lack of oxygen, it usually occurs in damp soils, sediments, and watercourses. At the moment, AD is the only biological treatment technique capable of producing biogas, a blend of methane (CH₄) and carbon dioxide (CO₂). By allowing organic materials to biodegrade in the absence of oxygen, AD replaces composting and produces a digested product that may be used as soil fertilizer. The AD process typically occurs in a closed vessel. Before being utilized in ultimate applications, the generated biogas is placed in temporary storage. The main commercial applications of biogas are the production of heat and

electricity. After AD, the jar will contain digestate, or residual solids and organic compounds. Digestate can be separated into liquid and solid streams. Because both streams contain vital plant nutrients, they can be applied as fertilizer in agricultural contexts[3]. Physical-chemical, biochemical, and microbiological processes are all part of the dynamic, complex AD process. Because it keeps biodegradable materials out of landfills and permits the creation of bioenergy and other byproducts like soil biofertilizer, AD has been determined to be the most environmentally benign method of managing biowaste[4]. The AD process involves a number of phases and intricate interactions between several microbes. At various phases, diverse microbial populations work together to break down the complex biomass polymers and transform them into a gaseous mixture. At particular stages, some bacterial groupings are more active than others. One stage's output serves as the input for subsequent stages. Kyoto Protocol to support biogas manufacturing as a renewable energy source. Much of the manufacturing and farm waste is perfect substrates for anaerobic digestion because they constitute elevated biodegradables. Pognani et al. (2009) describe the procedure in large-scale AD plants that produce biogas by blending energy crops with highly organic municipal solid waste. The methane ability of strong organic waste was evaluated in 2004 based on the type of organic substrate that influences the biodegradation processes and methane production, according to Forster-Carneiro et al. (2006) for effective uses of the AD method.

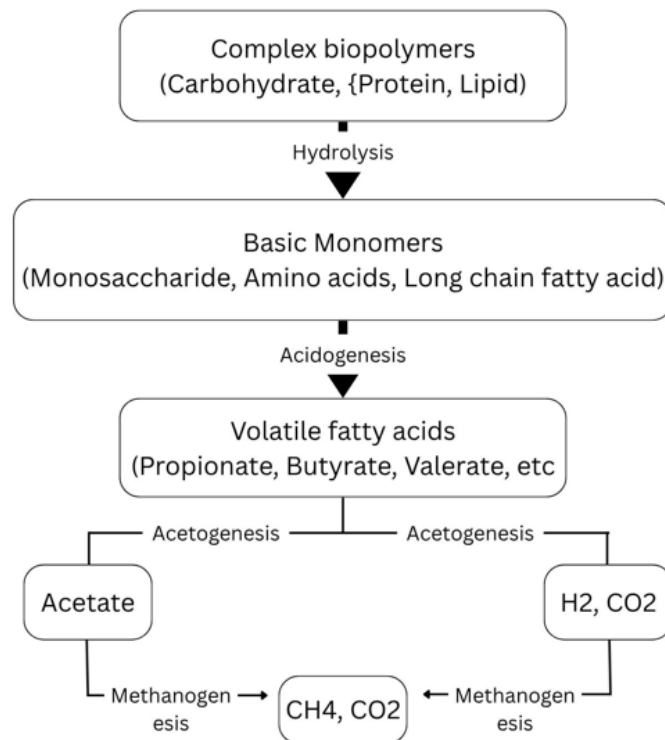


Figure 3.2(B): The Anaerobic Digestion pathways

Numerous metabolic connections between several microorganisms lead to AD. Anaerobic digestion occurs in three stages: hydrolysis, acidogenesis, and methanogenesis.

The first group of microbes release enzymes that hydrolyse polymers into monomers like sugar and amino acids. The second group, known as acetogenic bacteria, subsequently converts these into more volatile fatty acids, H₂ and acetic acid. Third, methanogenic bacteria produce methane from acetate, CO₂, and H.

Pre-treatment, substrate breakdown, gas regeneration, and manure extraction are the four main stages of the AD process. Most digesting methods require disposal pre-processing in order to obtain homogeneous material. Non-digestible goods must be separated and shredded as part of pre-processing. AD digesters often mechanically sort or separate the product from the source. Unwanted items like glass, metals, and stones are eliminated thanks to the separation. Recyclables and organic trash are segregated at the source. Mechanical segregation may be employed in the absence of source separation. Nevertheless, the final portion is further contaminated, which lowers the quality of the compost (Regional Information Service Centre for South East Asia on Appropriate Technology, 1998).

3.3 Substrates and Co-substrates

The majority of the material, such as animal dung, sewage sludge, or maize silage, is provided by the substrate, which acts as the main organic fuel for biogas production through Anaerobic Digestion (AD) and whose energy potential is determined by its Volatile Solids (VS) concentration. Substrates by themselves frequently have an unbalanced nutrient profile, most notably an unfavourable Carbon-to-Nitrogen (C/N) ratio (manures, for example, are overly rich in nitrogen), which can block the microorganisms that produce methane and cause hazardous ammonia to accumulate. In a procedure known as Anaerobic Co-Digestion (AcoD), a co-substrate an additional material such as food waste or fats, oils, and greases (FOG) is added to counteract this and maximize efficiency. By raising the overall C/N ratio to the ideal range of 20:1 to 30:1, supplying the required trace elements, and adding high-energy organic compounds, the co-substrate's job is to scientifically adjust the feed mix. This synergistic effect greatly raises the final methane content and guarantees the digester environment's long-term stability.

Several substrates and co-substrates are utilized for the production of biogas, including the following:

3.3.1 Primary Feedstocks:

- a. **Press Mud (Sugarcane Filter Cake):** A solid residue produced in sugar mills during the filtration of sugarcane juice is called press mud. It is mostly employed as a substrate, however because of its unique properties, co-digestion is frequently necessary. When co-digested with liquid waste, it can also act as a bulking agent.
- b. **Distillery Spent Wash (Vinasse/Stillage):** The liquid residue left over after fermented mash (often from grain or molasses) is distilled to make alcohol is known as distillery spent wash (DSW). Because of its high pollutant load, this high-strength liquid substrate is typically treated mostly using AD.
- c. **Cattle Manure / Slurry:** It is a conventional, high-volume base substrate that is rich in water and nitrogen (low C/N ratio), which facilitates mixing. Although it has a relatively modest gas production, it offers steady operation.
- d. **Sewage Sludge:** The solid wastewater treatment plant residue is a steady, dependable source. Because of its low organic content, it has a moderate to low biogas output and needs to be co-digested in order to recover energy.

- e. **Energy Crops (e.g., Maize Silage):** crops cultivated especially for AD; they have a very high biogas generation and are high in readily digested carbohydrates (starch). is frequently utilized as a co-substrate to improve manure effectiveness, but it requires dedicated land.
- f. **Agricultural Residues (e.g., Straw, Grass):** Husk and old grass are examples of by-products that break down slowly because they are frequently lignocellulosic (fibrous). Although they can have a high carbon content, their complex structure necessitates pre-treatment or co-digestion.

3.3.2 Co-Substrates (Performance Enhancers):

- a. **Food Waste / Kitchen Waste:** It has a very high potential for biogas, mostly because it contains a lot of easily digested carbs and lipids. To improve energy and adjust the C/N ratio, it is often co-digested with dung.
- b. **Slaughterhouse Waste (e.g., Blood, Paunch):** It is rich in lipids and protein, it has a high energy content and produces a good amount of biogas. Because of the high nitrogen/ammonia emission and possible infection danger, it needs to be managed carefully.

3.3.3 Biogas generation from Raw Spent Wash (Vinasse)

The wastewater generated by molasses-based distilleries, known as raw spent wash (RSW), has a very high organic burden, with COD levels usually ranging from 80,000 to 150,000 mg/L and BOD levels approaching 96,000 mg/L. Melanoidin pigments give it a dark brown hue. Polysaccharides, lignin, proteins, suspended particles, and a variety of micronutrients are present in this effluent, which is typically acidic (pH 3.8–5.5) and released at high temperatures. Anaerobic digestion can eliminate up to 85% of the COD from RSW during treatment, according to several studies.

The Characteristics table of Raw Spent Wash (Vinasse) is shown in the below table no: 1

SR. No.	Colour	Total Dissolved Solids(mg/l)	Total Suspended Solids(mg/l)	pH	COD	BOD	References
1	Dark brown	95000-110000	30000-40000	3.8-4.0	80000-100000	40000-50000	[29]

2	Dark brown	-	2400-5000	4.0-4.5	100000-150000	35000-50000	[30]
3	Dark brown	112400	27860	3.8	162000		[31]
4	Dark brown	58000-76000	2000-14000	3.8-4.4	70000-98000	45000-60000	[32]
5	Dark brown	-	-	3.9-4.3	104000-134400	46100-96000	[28]
6	Dark brown	-	10000	4-4.5	8000-120000	45000-60000	[33]
7	Dark brown	12600-14700	2100-6500	4.5-5.5	103000-158000	65000-92000	[34]

Table 3.3.3: Characteristics of RSW

A study performed by Manoj wagh et al. [46], looked at how well the OPUR treatment method, which uses anaerobic digestion to treat distillery wastewater (spent wash), worked in the climate of India. The primary findings were: The Chemical Oxygen Demand (COD) was approximately reduced from 140,000 to 38,500 mg L^{-1} which is a 65% to 72% reduction. The Biochemical Oxygen Demand (BOD) decreased by 80% to 85%, from 70,000 to 13,000 mg L^{-1} . The total dissolved solids (TDS) decreased by 75%, from 100,000 to 25,000. It was discovered that the particular biogas generation rate ranged from 0.45 to 0.55 Nm^3 per kilogram of COD decomposed [46]. A Study conducted by author Abdullah et al. [47], The analysis performed, focused on the anaerobic digestion of the spent wash at the Shakarganj Sugar Mills biogas power plant. So based on their results the findings were: Over the course of a year (October to September), 20.34 million M^3 of biogas were generated from 0.5 million M^3 of wasted wash. 55,760 M^3 of biogas were produced on average each day from 1,394 M^3 of spent wash. During the production of biogas, the Chemical Oxygen Demand (COD) decreased from a peak of 95,000 mg/l to 18,000 mg/l. The overall efficiency of COD reduction reached 81%. The system's efficiency in reducing Biochemical Oxygen Demand (BOD) was 83%. The Total Dissolved Solids (TDS) decreased by 44%. [47]. A Study conducted by Khaliyah Sania, Anun Salaeh, et al. [48] on the use of a Step Anaerobic Baffled (SAB) reactor for treating high-strength distillery spent wash and recovering biogas. The results from the study were: The Optimal S/I Ratio of 1:4 was found to

produce the highest methane yield and minimal inhibition. The COD removal efficiency was 85.4%. The Specific Methane Yield found was $299.0 \pm 13.5 \text{ mL } -CH_4/\text{g} - \text{COD}$. The Average daily Methane Production was $22.2 \pm 0.9 \text{ L/day}$ [48]. A Study was conducted by R. Shyam Sundar, R. Saravanan, et al., in which the study used an experimental model to use anaerobic digestion to produce and upgrade biogas from distillery waste wash. The Experimental Observations were: The biogas produced from the spent wash showed the following chemical characteristics: The Methane content was found to be 50-75%. The Carbon Dioxide was 25-50%. The hydrogen Sulphide was 0-3%. The Nitrogen content was 0-10%. The hydrogen content was 0-1%.

3.3.4 Biogas generation from Press Mud (Filter Cake)

When sugarcane juice is clarified, press mud, also called filter cake, is produced. It has a high percentage of organic matter, lignocellulosic chemicals, waxes, proteins, and minerals in addition to 60–85% moisture. Co-digesting it with carbon-rich materials can enhance its carbon-to-nitrogen (C/N) ratio, which typically ranges from 9 to 14. [49][43][50].

Sr No.	Moisture (%)	TS (%)	VS (%)	Lignin (%)	Cellulose (%)	Hemicellulose (%)	C/N Ratio	pH	References
1	-	47.850 ± 0.002	34.780 ± 0.013	-	-	-	9.682	-	[51]
2	57.54	-	-	-	-	-	9.5	7.56	[52]
3				17.4	12.8	6.5	-	-	[53]
4	-	-	-	9-14	1-17	15-27	-	-	[54]
5	76.3	-	-	9.3	11.4	10.0	14.0	7.5	[55]
6		27.7	80.8	9.3	11.3	27.1	26:1	6.3	[56]
7	77.8	-	-	9.0	11.6	10.2	14.2	7.1	[57]

Table 3.3.4: Characteristics of Press Mud (Filter Cake)

A Study was conducted by M. A. Rouf, P. K. Bajpai et al. [55]. The study looked into how to best produce biogas from press mud in a batch reactor, giving following results: With press mud with 6% solid concentration, the maximum biogas yields ever measured was 0.24 L/g. It was discovered that 18 was the ideal Carbon/Nitrogen (C/N) ratio for producing biogas. This is greater than the typical press mud (C/N) ratio of about 14. When cane pith and press mud were combined in a 1:1 ratio, the amount of biogas produced increased dramatically. The final biogas that was produced contained approximately 65% of Methane [55]. A similar Study was performed by M. A. Rouf, M. S. Islam, et al [57]. The study evaluated the technical and financial feasibility of turning press mud into biogas. The following results were: A total gas production of 0.18 L/g of press mud was attained. The biogas that was produced contained 65% of Methane. A carbon-to-nitrogen (C/N) ratio of 13 to 15 was found, which is just below the typical threshold for ideal bio methanation. The press mud sample that was used had a considerable amount of volatile matter that is 75-80%. In a 150 cft (cubic feet) facility, biogas production was investigated at solid concentrations ranging from 6% to 8% [57]. A Study was conducted by Srinivas Kasulla et al. [58] The main conclusion of the research is that adding small components (Iron, Nickel, and Cobalt) is one of the greatest ways to increase press mud biogas output by 20–30%. The following results were: A final methane output of 430–460 l/kg of organic dry matter was obtained. The amount of crude biogas produced per ton of press mud after conversion was between 82 and 90

M^3 . In certain instances, the generation of crude biogas was close to 100–105 M^3 per ton of press mud. When nickel and cobalt were added, biogas production started the quickest (from the very first day). When press mud was processed anaerobically, the inclusion of iron produced the highest total yield [58].

3.3.5 Co-digestion

The concept of co-digestion developed more gradually when researchers and operators started experimenting with combining various organic wastes to increase gas yields and process stability, despite the fact that anaerobic digestion had been used for more than a century. The advantages of mixing substrates became more apparent as waste streams grew and technology developed, but early digesting systems usually only handled one substance at a time.

The breakdown of a homogeneous mixture consisting of two or more substrates that are digested sequentially is referred to as co-digestion. Large amounts of a primary substrate, such manure or sewage sludge, are typically mixed with lesser amounts of one or more additional elements. Regardless of the relative amounts of each component in the mixture, the term "co-digestion" is applicable. In the past, anaerobic digestion was a single-substrate, single-purpose process that produced biogas primarily from manure, stabilized sewage sludge anaerobically, and pretreated industrial effluent prior to ultimate treatment. As knowledge of the process increased, co-digestion's benefits and adaptability became more apparent, resulting in its widespread use. Modern agriculture demands high-quality, contaminant-free fertilisers, co-digestion is well suited to combining carefully monitored industrial wastes, park grass clippings, food industry residues, dairy waste, and similar materials with dilute substrates like sewage sludge or animal manure [59].

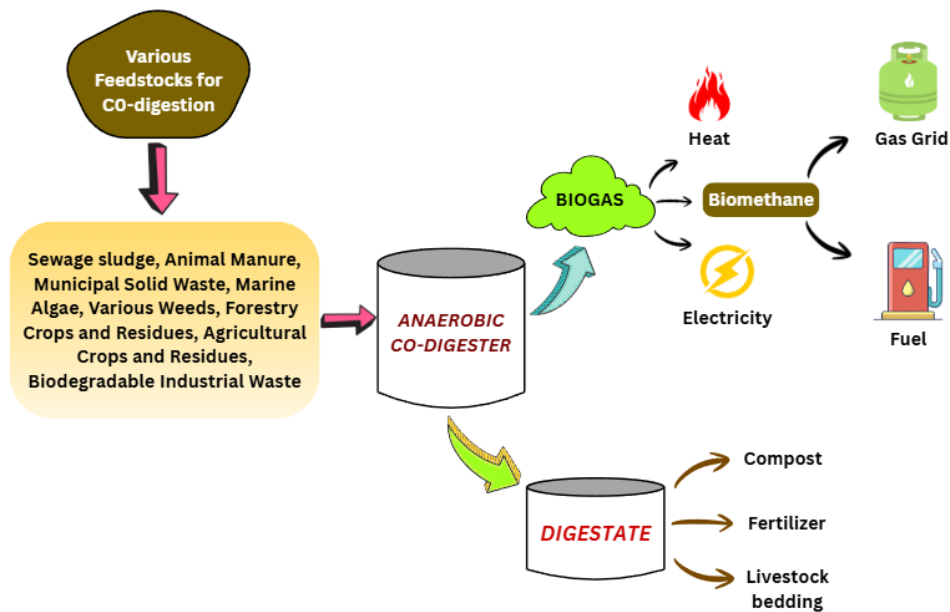


Figure 3.3.5: Co-digestion of multiple feedstocks for waste reduction and energy recovery

3.3.5.1 There are two major drivers which helped to promote co-digestion:

1. In wastewater treatment facilities, digesters are frequently constructed larger than what is required for the sludge they typically handle. With only a slight increase in operational costs, these systems may produce more biogas and, consequently, more electricity by incorporating more substrates. The additional electricity generated can greatly offset the treatment plant's energy requirements [59].
2. Similarly, with current fuel costs, it is not financially feasible to produce biogas on farms using simply manure, which has a relatively low biogas yield. In addition to increasing overall gas output, combining manure with high-methane-yield materials produces additional income through tipping fees earned for accepting those extra wastes [59].

3.3.5.2 Merits of Co-digestion

1. Enhances Biogas Yield (Synergistic Effect): By combining nitrogen-rich vinasse with carbon-rich press mud, co-digestion maximizes the C:N ratio (optimal ~20 or 30:1). When compared with mono-digestion of either substrate, this synergy enhances methane generation. By increasing buffering capacity, press mud lowers the possibility of vinasse

acidification. In typical sugarcane distillery systems, co-digestion increases biogas generation by 15–40% [60][61][62].

2. Improves Process Stability of the Digester: Vinasse can lead to the buildup of volatile fatty acids (VFAs), has a low pH, and has a high organic load (COD 80,000–120,000 mg/L). CaCO_3 , fibre, and minerals included in press mud aid in pH buffering and inhibitor dilution. Co-digestion balances content of moisture, Rate of organic loading (OLR), Ratios of nutrients (C: N:P:S). Reduces the shock load on microorganisms and ensures stable methanogenesis [60][63].

3. Increases Biodegradability and Organic Matter Utilization: Press mud has a lot of lignocellulosic chemicals, while vinasse offers organic acids and sugars that are easily broken down. Complementary microbial pathways are made possible via co-digestion, leading to: Quicker hydrolysis, Improved acidogenesis, Effective conversion of methanogen. Leads to higher reduction in COD, BOD, VS, improving overall system efficiency [61][63].

4. Enhanced Nutrient Recycling Through Digestate: A nutrient-rich digestate produced by co-digestion is suitable for use as an organic fertilizer. Vinasse provides potassium, nitrogen, and sulphur, while press mud adds calcium, phosphorus, and other important micronutrients. Combining these residues results in a digestate that has a balanced N-P-K profile, increases humus formation, and improves soil conditioning and water-holding capacity. By returning important nutrients to agricultural fields, this lessens reliance on artificial fertilizers and promotes a circular economy within sugarcane distillery operations [64][61].

5. Improved Energy Recovery and Economic Returns:

By raising the total amount of methane produced, increasing the calorific value of the biogas, and increasing the amount of energy recoverable per ton of processed residue, co-digestion improves the system's overall energy potential. Economically, this results in a number of benefits, such as reduced waste treatment costs, increased energy generation from increased biogas output, lower effluent handling costs, and more revenue potential from the sale of nutrient-rich digestate as a high-value biofertilizer [61][65].

6. Waste Management and Environmental Benefits: By effectively digesting vinasse, a highly polluting and challenging-to-manage wastewater, co-digestion reduces environmental contamination and removes the need for press mud to be dumped outdoors. By capturing methane that would otherwise escape from unmanaged wastes, the technique lowers greenhouse gas emissions. It also helps achieve zero-liquid-discharge goals and encourages more environmentally friendly and sustainable distillery operations [64][66].

3.3.5.3 Co-Digestion of Press Mud with other Substrates

Various studies have been conducted on Co-digestion of Press Mud with other Substrates like Press Mud with Vegetable waste, Abattoir wastewater, Food waste, Vinasse (RSW) and Distillery Wastewater, etc. Below table III. Shows the percentage increased in the CH₄ yield co-digesting with different substrates.

Co-digestion material	Conditions	Increased CH ₄ Yield (%)	References
Vinasse	Press mud and Vinasse in the ratio 50:50	12-13%	[60]
Vegetable Waste	Press mud and vegetable waste in the ratio 30:70	31%	[67]
Abattoir wastewater	Sugarcane Press mud and Abattoir wastewater in ratio 80:20	141%	[68]
Distillery wastewater+ nutrients	Press mud and Distillery wastewater in the ratio 1:2	76%	[69]
Food Waste	Sugarcane Press mud and Food waste in the ratio 80:20	22%	[70]

Table 3.3.5.3: Co-Digestion of Press Mud with other Substrates

Chapter 4

METHODOLOGY

4. MATERIALS AND METHODOLOGY

4.1 Materials

4.1.1 Introduction

This chapter focuses on the evaluation of key operational parameters such as Total Solids (TS), Volatile Solids (VS), Moisture Content (MC), the ratio of Volatile Solids to Total Solids (VS/TS), and Chemical Oxygen Demand (COD). It describes the analytical procedures and instruments used for the determination of these selected parameters. Furthermore, the chapter explains the methodology adopted for evaluating the Biochemical Methane Potential (BMP) of the substrates using the AMPTS-III instrument. In addition, the procedure employed to estimate the projected BMP for the co-digestion of PM and RSW substrates at different mixing ratios is also presented.

4.1.2 Press Mud (Filter Cake)

Fresh press mud obtained from a local sugar industry was utilised as a substrate for co-digestion in the present study to investigate its potential for biomethanation. Press mud is considered a suitable feedstock due to its high moisture content and significant organic matter, which enhance microbial activity and support efficient biogas production. Co-digestion of press mud with other biodegradable substrates can further improve methane generation and overall process efficiency in biogas plants. Therefore, RSW was selected as the co-substrate for the co-digestion experiments in this study. The Press Mud sample was then stored at 4°C to minimise microbial activity prior to experimentation. Before conducting the Biochemical Methane Potential (BMP) test, the PM was brought to room temperature and thoroughly mixed to obtain a homogeneous and representative sample.



Figure 1.1.2: Press Mud

4.1.3 Raw Spent Wash (Vinasse)

Raw Spent Wash (RSW) used in the present study was collected from a nearby distillery unit associated with a local sugar industry. The collected sample was transported to the laboratory in airtight containers to avoid contamination and degradation. Prior to experimental analysis, the RSW underwent preliminary treatment, as the collected sample was initially B-heavy molasses based spent wash with COD of 84,000, which was subsequently concentrated (evaporated at 80°C) and converted into C-heavy molasses based spent wash as the required COD for this study was around 1,44,000. The sample was then stored at 4°C to minimize microbial activity prior to experimentation. Before conducting the Biochemical Methane Potential (BMP) test, the RSW was brought to room temperature and thoroughly mixed to obtain a homogeneous sample. The prepared RSW was subsequently used as a co-substrate with press mud for the co-digestion studies carried out using the AMPTS-III system.



Figure 4.1.3: Raw Spent Wash (Vinasse)

4.1.4 Equipment Used

The study involves extensive equipment usage to measure biogas potential using a BMP instrument. This investigation focuses on the bio-methanation of Press Mud and Raw Spent Wash under BMP conditions. The equipment utilised in this project is listed below:

4.1.4.1 Biochemical Methane Potential (BMP) Instrument

A BMP device enables the determination of a biomass substrates dynamic degradation profile as well as its actual biogas and methane potential. This enables the effective screening for suitable pre-treatment techniques, the recognition of ideal retention periods and substrate combinations for co-digestion, the evaluation of the biological

performance of specific digesters or process configurations, and the determination of the necessity for additives.

The **Automatic Methane Potential Test System (AMPTS) BMP instrument** is a specialized system designed to measure the Biochemical Methane Potential (BMP) of organic substrates subjected to anaerobic digestion. It provides accurate and completely automated measures of methane generation by tracking gas output using various feedstocks. The BMP system maintains ideal temperature, pressure, and mixing conditions throughout digestion to guarantee accurate and high-quality data. It is designed for both industrial and research applications. It is frequently used to measure methane yields from various organic materials in waste management, renewable energy research, and biogas production studies. This assists with substrate selection and biogas plant performance optimization.

In the first stage of the process, the inoculum and substrate are incubated under anaerobic conditions. A thermostatic water bath is used to maintain and control the incubation temperature at 38°C. During this stage, biogas composed primarily of carbon dioxide (CO₂) and methane (CH₄) is generated. In the second stage, the produced biogas passes through the absorption unit, where carbon dioxide is absorbed by an alkaline sodium hydroxide (NaOH) solution, allowing methane to remain in the gas stream. In the third stage, the methane gas enters the detection unit through a flow cell where it is quantitatively measured. The measured data are recorded and stored in the BPC AMPTS software system, which can be accessed through a connected computer or via an internet interface.

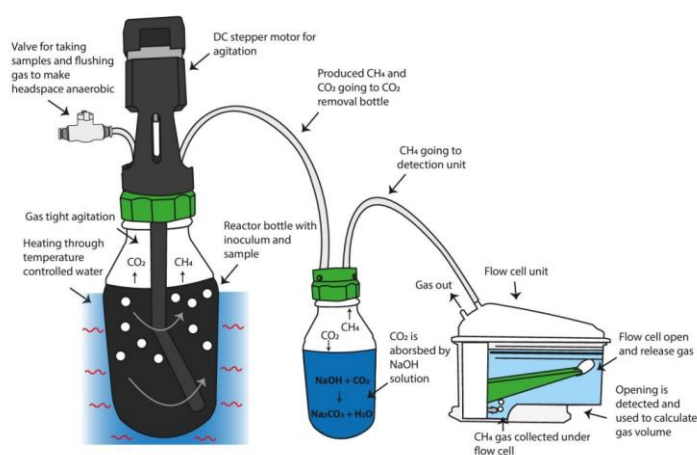


Figure 4.1.4.1: Working of BMP Instrument



Figure 4.1.4.1: AMPTS Instrument

4.1.4.2 Hot Air Oven

A hot air oven is a piece of equipment used in laboratories that uses dry heat to sterilize, dry, and treat materials. In order to eliminate moisture and kill microbes, it circulates hot air at regulated temperatures, usually between 50°C and 300°C.



Figure 4.1.4.2: Hot Air Oven

4.1.4.3 Muffle Furnace

A muffle furnace is a high-temperature laboratory device which utilizes an insulated chamber for the ashing, burning, melting, and heat treatment of materials. It can function at temperatures between 300°C and 1800°C, depending on the model and use. The furnace ensures consistent and regulated heat distribution by keeping the sample away from the heating elements.



Figure 4.1.4.3: Muffle Furnace

4.1.4.4 Other Equipment and material used

1. pH Meter
2. Weighing Balance
3. Centrifuge Machine
4. Desiccator
5. COD Digester



Figure 4.1.4.4: pH Meter



Figure 4.1.4.4: Weighing Balance



Figure 4.1.4.4: Centrifuge Machine

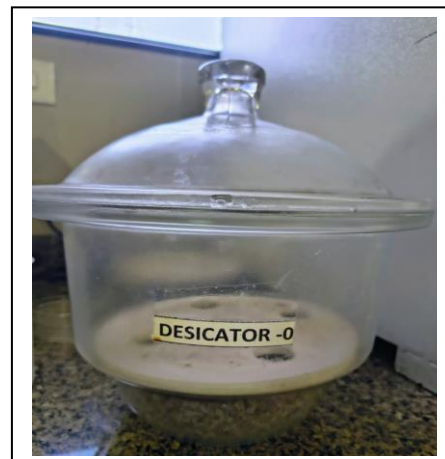


Figure 4.1.4.4: Desiccator



Figure 4.1.4.4: COD
Digester

4.2 Methodology

4.2.1 Mixing Ratio of Co-Digestion of Press Mud and Raw Spent Wash

Different mixing ratios of Press mud and Raw spent wash used for Biochemical Methane Potential (BMP) analysis are as follows:

Sr. No.	Mixing Ratio (Press mud: Raw spent wash)
1	RSW
2	PM
3	1:2
4	1:3

Table 4.2.1: Mixing ratios of PM & RSW

4.2.2 Characterization

4.2.2.1 Total Solids (TS)

Using a pre-weighed dish, a well-mixed sample is evaporated and dried to a constant weight in an oven set at 103–105°C. The residue that remains after drying represents the substrate's overall solids content. The weight of the residue in relation to the wet sample is then used to compute the total solids.

$$\% \text{ Total Solids} = \frac{A - B}{C - B} * 100$$

where,

A = weight of dried residue + dish (g)

B = weight of dish (g)

C = weight of wet sample + dish (g)

D = weight of residue + dish after ignition (g)

4.2.2.2 Volatile Solids (VS)

The residue obtained from the total solid's determination is ignited at 550°C until a constant weight is achieved. The volatile solids are represented by the weight loss during ignition, and the fixed, dissolved, or suspended inorganic components are represented by the residual ash. The standard formula is then used to compute total volatile solids. This value is crucial because it provides an accurate indication of the amount of organic matter in the waste, which is necessary for evaluating and controlling biological treatment procedures.

$$\% \text{ Volatile Solids} = \frac{A - D}{A - B} * 100$$

A = Weight of dried residue + dish (g)

B = Weight of dish

D = Weight of residue + dish after ignition (g)

3.2.2.3 Moisture Content (MC)

The weight loss in an Oven set to 103°C to 105°C after the samples have been dried is the moisture content. The following calculation of moisture content is based on the wet basis.

$$\% \text{ Moisture Content} = \frac{w - d}{w} * 100$$

Where,

w = Initial (wet) weight of sample 1/3rd

D = Final (dry) weight of sample

4.2.2.4 Total Dissolved Solids (TDS)

Total Dissolved Solids (TDS) represent the fraction of solids that pass through a standard glass fibre filter and remain in the filtrate. Organic matter, dissolved inorganic salts, and other soluble materials found in the substrate make up the majority of TDS. To fully understand the material's dissolved load, which affects anaerobic digestion efficiency and process stability, TDS measurement is crucial.

For TDS analysis, a well-mixed sample is first filtered through a standard glass fiber filter to remove suspended solids. Following that, a predetermined volume of the filtrate is put into an evaporating dish and dried at 103–105°C in a hot air oven until a constant weight is achieved. The total amount of dissolved solids in the sample is indicated by the rise in dish weight during drying.

$$\% \text{ Total Dissolved Solids (TDS)} = \frac{w_1 - w_2}{V} \times 1000$$

Where,

w₁ = Weight of dried residue + dish (mg)

w₂ = Weight of empty dish (mg)

V = Volume of sample filtered (mL)

4.2.2.5 Total Suspended Solids (TSS)

Total Suspended Solids (TSS) represent the fraction of solids that are retained on a standard glass fiber filter and are not dissolved in the liquid phase. Organic particles and inorganic matter make up the majority of suspended solids, which have an important influence on the substrate's biodegradability and digestion kinetics during anaerobic treatment.

For TSS determination, a known volume of a well-mixed sample is filtered through a pre-weighed glass fiber filter. Following that, the filter with the residual solids is dried at 103–105°C in an oven until it reaches a consistent weight. The amount of suspended particles in the sample is indicated by the increase in filter weight during drying.

$$\% \text{ Total Suspended Solids (TSS)} = \frac{w_1 - w_2}{V} \times 1000$$

Where,

w_1 = Weight of dried filter + Residue (mg)

w_2 = Weight of clean filter paper (mg)

V = Volume of sample filtered (mg)

4.2.3 Chemical Analysis

4.2.3.1 Chemical Oxygen Demand (COD)

Chemical Oxygen Demand (COD) is a measure of the amount of oxygen required to chemically oxidize the organic and inorganic matter present in the substrate. Since COD gives an estimate of the overall biodegradable and non-biodegradable organic load, it is an essential indicator for assessing the material's organic strength and suitability for anaerobic digestion.

The COD of the sample was calculated using the following equation,

$$\text{COD (mg/L)} = \frac{B - B.R \times 0.1N \times 8000}{\text{Vol of Sample}}$$

Where,

B = blank

$B.R$ = Burette Reading

4.2.4 Biochemical Methane Potential

The Biochemical Methane Potential (BMP) or Biogas Potential Test assesses the potential amount of methane or biogas produced by a substrate.

The test comprises two main components: (a) the inoculum and (b) the substrate sample typically characterized in terms of volatile solids (VS) or COD. The combined inoculum (50:50 PM and RSW digester outlet) was used because it contains a pre-acclimatized and diverse microbial culture adapted to both substrates, thereby enhancing process stability, reducing lag phase and improving methane yield. A defined quantity of inoculum is added to each reactor, followed by a measured amount of substrate based on a specified inoculum-to- substrate ratio (ISR) determined by VS content. The measured biomethane volume and flow are normalised under standard condition (i.e., 0 °C, 1 atm and zero moisture) and the measurement interference caused by variations of external measurement conditions can be minimised.

The biogas produced by the inoculum is subtracted from the total biogas generated in each test bottle to determine the net biogas production attributable to the substrate's gVS in that bottle.

The Co-digestion mixing ratios of Press Mud and Raw Spent Wash were set at RSW, PM, (1PM: 2RSW), (1PM: 3RSW). All reactors were mixed at identical speeds to ensure uniform and consistent mixing.

After the results of Biochemical Methane Potential, we calculate the targeted BMP values of mixing ratio of 1:2 (PM: RSW) and 1:3 (PM: RSW) to compare with the actual BMP generated from these ratios.

For calculation of BMP,

$$\text{BMP} = \frac{\text{Total gas produced (Inoculum + Substrate)} - \text{Total gas produced (Inoculum)}}{\text{VS in grams of Substrate}}$$

For the calculation of projected BMP, we multiply the actual BMP of Press Mud and Raw Spent Wash by the multiplying factor of the mixing ratio.

For Projected BMP value of 1:2 (PM: RSW)

$$= \frac{1 \times \text{Actual value of BMP of PM} + 2 \times \text{Actual value of BMP of RSW}}{3}$$

For Projected BMP value of 1:3 (PM: RSW)

$$= \frac{1 \times \text{Actual value of BMP of PM} + 3 \times \text{Actual value of BMP of RSW}}{4}$$

Now, after calculation of the projected BMP values, we calculate the actual increase in the BMP

Actual Increase in the BMP value of 1:2 (PM: RSW)

$$= \text{Actual value of BMP} - \text{Targeted value of BMP}$$

Actual Increase in the BMP value of 1:3 (PM: RSW)

$$= \text{Actual value of BMP} - \text{projected value of BMP}$$

Based on the observed increase in the BMP values, the optimum mixing ratio for co-digestion of press mud and Raw Spent Wash was identified.

4.2.5 Assessment of Fertilizer Reduction Potential using Biomethanated Spent Wash and Press Mud

Apart from the biochemical methane potential (BMP) analysis, an additional assessment was conducted to evaluate the fertilizer reduction potential of biomethanated spent wash (BMSW) and press mud in sugarcane cultivation. The study involved analysing the nutrient contribution of these by-products and estimating their potential to partially replace conventional chemical fertilizers such as Diammonium Phosphate (DAP) and urea. For this purpose, the nutrient composition of biomethanated spent wash and press mud, particularly nitrogen (N) and phosphorus (P) content, was considered. The conventional fertilizer requirement for sugarcane cultivation was taken as the baseline reference, where approximately 200 kg of DAP and 200 kg of urea per acre are generally applied in traditional practices. The methodology involved estimating the equivalent nutrient supply provided by the application of biomethanated spent wash at a rate of approximately 100 L per ton of sugarcane and press mud at about 40 kg per ton of sugarcane. Based on the nutrient content of these materials, their contribution to nitrogen and phosphorus availability was calculated and compared with the nutrient supply from conventional fertilizers. This comparative evaluation helped determine the extent to which chemical fertilizer requirements could be reduced through the utilization of biomethanated spent wash and press mud, thereby promoting resource recycling and sustainable nutrient management in sugarcane cultivation.

This analysis was carried out to understand the agricultural co-benefits of biomethanated spent wash and press mud utilization, complementing the energy recovery potential evaluated through the BMP study.

Chapter 5

RESULTS AND DISCUSSIONS

5. RESULTS AND DISCUSSIONS

5.1 SET 1 of Co-digestion of Press Mud & Raw Spent Wash

5.1.1 Characterisation of Press Mud and Raw Spent Wash

The initial phase of this investigation involved the characterisation of press mud and raw spent wash. B-heavy spent wash (RSW) with an initial COD of 84,000 ppm was subjected to evaporation in a hot air oven maintained at 80°C to obtain characteristics closer to C-heavy molasses, resulting in an increased COD. Table 4.1 presents the characteristics of both substrates. To improve accuracy and reliability, all parameters were analysed in duplicate.

Parameters	Unit of Measurement	Press Mud	Raw Spent Wash
Moisture content	%	71.59	85.89
Total Solids	%	28.42	14.51
Volatile Solids	%	20.2	10.93
Volatile Solids (% of TS)	%	71.10	75.29
pH	-	-	4.37
COD	mg/l	-	140,000
TDS	mg/l	-	1,27,400
TSS	mg/l	-	4,800

Table 5.1.1: Characterization of Substrates

The above table indicates a clear difference between press mud and raw spent wash in terms of physical and biochemical characteristics relevant to anaerobic digestion. Press mud, with lower moisture content (71.59%) and higher total solids (28.42%), is a semi-solid substrate rich in organic matter (VS = 20.2%, 71.10% of TS), making it suitable for stable digestion. In contrast, raw spent wash is a highly dilute yet extremely high-strength liquid waste (moisture 85.89%, COD 140,000 mg/L) with a slightly higher biodegradable fraction (VS = 75.29% of TS), indicating strong potential for rapid biogas production. Press mud provides bulk organic content and buffering capacity, while spent wash contributes readily degradable substrates, making their co-digestion an effective strategy to enhance methane yield, improve process stability, and optimize overall anaerobic digestion performance.

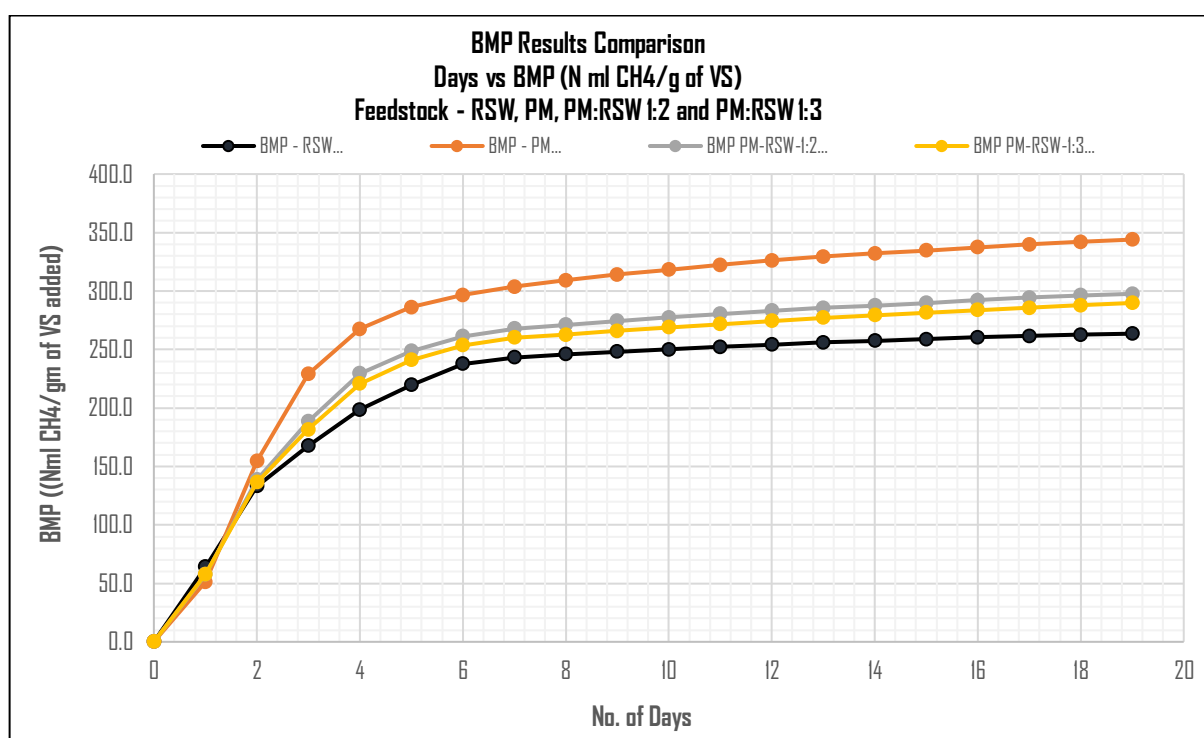
5.1.2 Biochemical Methane Potential of various mixing ratios of Press Mud and Raw Spent Wash.

The Biochemical Potential (BMP) is a measure of how much methane microbial activity can create from a particular organic substrate (such as food waste, manure, sewage sludge) under anaerobic conditions. Press Mud and Raw Spent Wash samples were fed into the BMP digesters alone and in a mixture of PM: RSW (1:2) and PM: RSW (1:3) for methane potential measurement at 38°C. The unit for BMP is Nml CH₄/gm VS added.

No. of Days	PM	RSW	1:2 (PM: RSW)	1:3 (PM: RSW)
	Nml CH ₄ /gm VS	Nml CH ₄ /gm VS	Nml CH ₄ /gm VS	Nml CH ₄ /gm VS
0	0.0	0.0	0.0	0.0
1	51.2	64.1	57.5	57.9
2	154.5	133.2	138.7	136.5
3	228.8	167.6	188.3	181.5
4	267.5	198.5	229.4	220.7
5	286.1	219.7	248.6	241.0
6	296.7	237.7	261.3	253.6
7	303.7	243.2	267.7	260.1
8	309.1	246.0	271.0	262.5
9	314.1	248.1	274.4	265.8
10	318.1	250.1	277.4	268.8
11	322.3	252.1	280.5	271.7
12	326.1	254.1	283.2	274.5
13	329.3	256.0	285.6	277.1
14	332.2	257.5	287.6	279.3
15	334.7	258.9	289.8	281.5
16	337.4	260.3	292.1	283.7
17	339.9	261.4	294.3	285.7
18	342.1	262.5	296.2	287.7
19	344.0	263.5	297.6	289.8

Table 5.1.2: BMP of various mixing ratios of PM & RSW

These results show that methane production increases in the initial days followed by a plateau phase after about 15–20 days, indicating completion of digestion. Press Mud (PM) exhibits a BMP ~340–345 N ml CH₄/g VS, while Raw Spent Wash (RSW) shows BMP ~260–270 mL CH₄/g VS. Co-digestion improves methane yield compared to RSW alone, demonstrating a synergistic effect. Among the mixtures, PM: RSW (1:2) performs better than (1:3). Overall, co-digestion had no significant negative effect but also did not provide substantial improvement.



Graph 5.1.2: Methane Generation of various Mixing ratios over time (SET-I)

Table II & III shows the comparison of actual and projected values of BMP for 1:2 & 1:3 ratios. The projected BMP values were determined by calculating the arithmetic average of individual BMP values.

❖ **Actual value of BMP and Projected value of BMP of ratio 1:2 (PM: RSW)**

No. of Days	Actual BMP	Projected BMP	% Increase in BMP
	Nml CH ₄ /gm VS	Nml CH ₄ /gm VS	Nml CH ₄ /gm VS
1	57.5	59.8	-3.85

2	138.7	140.3	-1.14
3	188.3	188.0	0.16
4	229.4	221.5	3.57
5	248.6	241.8	2.81
6	261.3	257.3	1.55
7	267.7	263.3	1.67
8	271.0	267.0	1.50
9	274.4	270.1	1.59
10	277.4	272.8	1.69
11	280.5	275.5	1.81
12	283.2	278.1	1.83
13	285.6	280.4	1.85
14	287.6	282.4	1.84
15	289.8	284.1	2.01
16	292.1	286.0	2.13
17	294.3	287.6	2.33
18	296.2	289.0	2.49
19	297.6	290.3	2.51

Table 5.1.2: Actual & Targeted BMP values of ratio 1:2

The results indicate that during the initial phase of digestion, the actual BMP values were slightly lower than the projected BMP values, resulting in a small negative percentage difference. However, as the digestion process progressed, the actual BMP gradually exceeded the projected BMP values, showing a positive percentage increase. This trend suggests that the microbial activity improved over time, enhancing methane generation during co-digestion of press mud and raw spent wash at a 1:2 ratio. The selected ratio resulted in only a marginal increase in BMP.

❖ **Actual value of BMP and Projected value of BMP of ratio 1:3 (PM: RSW)**

No. of Days	Actual BMP	Projected BMP	% Increase in BMP
	Nml CH ₄ /gm VS	Nml CH ₄ /gm VS	Nml CH ₄ /gm VS
1	57.9	60.9	-4.93
2	136.5	138.5	-1.44
3	181.5	182.9	0.77
4	220.7	215.7	2.32
5	241.0	236.3	1.99
6	253.6	252.4	0.48
7	260.1	258.3	0.70

8	262.5	261.7	0.31
9	265.8	264.6	0.45
10	268.8	267.1	0.64
11	271.7	269.7	0.74
12	274.5	272.1	0.88
13	277.1	274.3	1.02
14	279.3	276.1	1.16
15	281.5	277.8	1.33
16	283.7	279.6	1.47
17	285.7	281.1	1.64
18	287.7	282.4	1.88
19	289.8	283.6	2.19

Table 5.1.2: Actual & Targeted BMP values of ratio 1:3

Initially, the actual BMP values were slightly lower than the projected BMP values, showing a small negative percentage difference during the early digestion stage. As the digestion progressed, the actual BMP gradually exceeded the projected BMP values, resulting in a positive increase in methane potential.

5.2 SET 2 of Co-digestion of Press Mud & Raw Spent Wash

Immediately after the completion of Set 1, Set 2 was initiated following the same experimental procedure throughout. The same samples of PM and RSW used in SET 1 were also used in SET 2.

5.2.1 Characterization of Press Mud and Raw Spent Wash

Table 4.2.1 presents the Characterization of both the substrates for Set 2. To improve accuracy and reliability, all samples were analysed in duplicate.

Parameters	Units of Measurement	Press Mud	Raw Spent Wash
Moisture content	%	73.0	86.78
Total Solids	%	26.99	13.21
Volatile Solids	%	20.18	9.92
Volatile Solids (% of TS)	%	74.8	74.5
pH	-	-	4.37
COD	mg/l	-	140,000
TDS	mg/l	-	1,27,400
TSS	mg/l	-	4,800

Table 5.2.1: Characterisation of Substrates

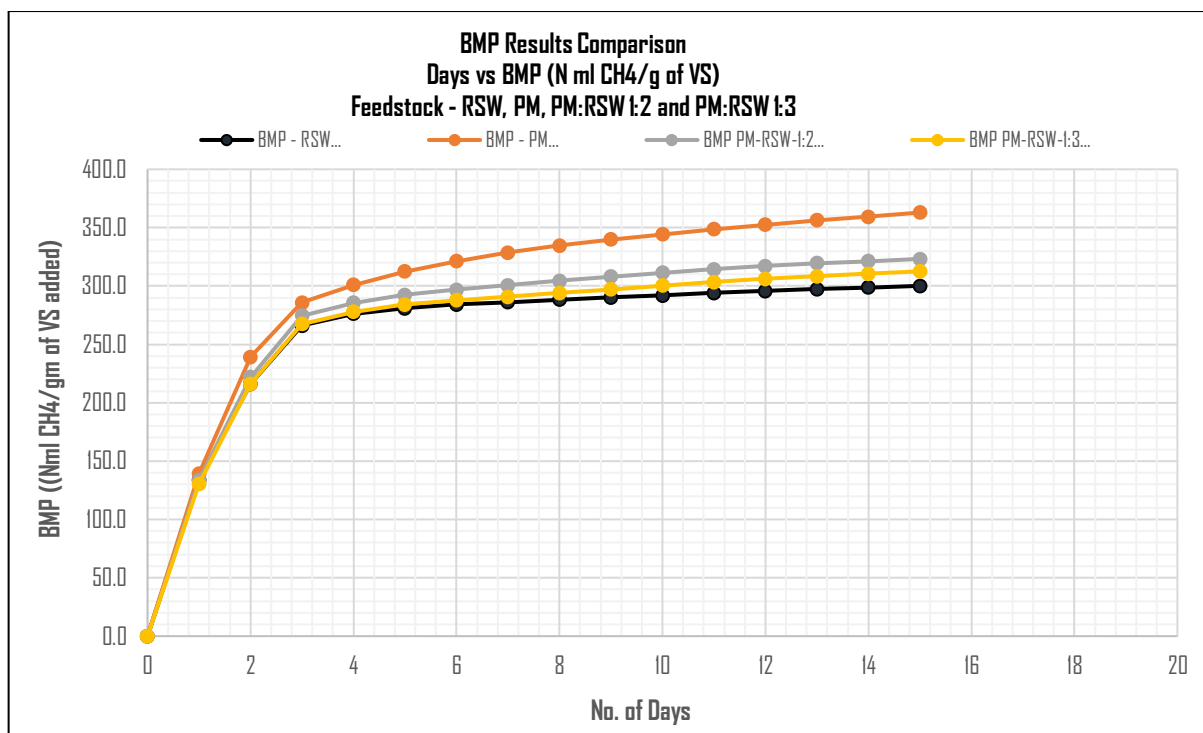
5.2.2 Biochemical Methane Potential of various mixing ratios of Press Mud and Raw Spent Wash.

Table 4 shows the BMP of different mixing ratios of Press Mud and Raw Spent Wash for SET 2, performed for 15 days.

No. of Days	PM	RSW	1:2 (PM: RSW)	1:3 (PM: RSW)
	Nml CH ₄ /gm VS	Nml CH ₄ /gm VS	Nml CH ₄ /gm VS	Nml CH ₄ /gm VS
0	0.0	0.0	0.0	0.0
1	139.2	133.9	133.6	130.3
2	239.1	216.1	221.9	216.2
3	285.8	266.1	274.6	267.6
4	301.1	276.3	285.5	277.8
5	312.4	281.0	292.4	284.2
6	321.3	284.2	296.9	287.8
7	328.5	286.1	300.8	290.9
8	334.6	288.2	304.5	294.2
9	339.8	290.1	307.9	297.2
10	344.4	292.0	311.3	300.3
11	348.7	294.1	314.5	303.5
12	352.6	295.8	317.2	306.1
13	356.2	297.4	319.6	308.4
14	359.5	298.7	321.2	310.4
15	362.9	300.2	323.3	312.6

Table 5.2.2: BMP of various mixing ratios of PM & RSW

The table indicates that methane production increased during the initial days and stabilized after around 15 days. The BMP for Press Mud (PM) was observed to be approximately 360–365 N ml CH₄/g VS. The co-digestion ratios PM:RSW (1:2) and PM: RSW (1:3) produced yields of about ~320–325 and 310–315 N ml CH₄/g VS, respectively. In comparison, Raw Spent Wash (RSW) showed the BMP values of around 295–300 N ml CH₄/g VS. Overall, co-digestion improved methane production compared to RSW alone, with 1:2 ratio performing better than 1:3.



Graph 5.2.2: Methane Generation of various Mixing ratios (SET-II)

Table II & III shows the comparison of actual and projected values of BMP for 1:2 & 1:3 ratios. The projected BMP values were determined by calculating the arithmetic average of individual BMP values.

❖ **Actual value of BMP and Projected value of BMP of ratio 1:2 (PM: RSW)**

No. of Days	Actual BMP	Projected BMP	% Increase in BMP
	Nml CH ₄ /gm VS	Nml CH ₄ /gm VS	Nml CH ₄ /gm VS
1	133.6	135.7	-1.55
2	221.9	223.8	-0.85
3	274.6	272.7	0.70
4	285.5	284.5	0.35
5	292.4	291.4	0.34
6	296.9	296.5	0.13
7	300.8	300.2	0.20
8	304.5	303.7	0.26
9	307.9	306.7	0.39
10	311.3	309.5	0.58
11	314.5	312.3	0.70

12	317.2	314.8	0.76
13	319.6	317.0	0.82
14	321.2	319.0	0.69

Table 5.2.2: Actual value & Targeted value of BMP of ratio 1:2

The results show that the actual BMP values are slightly higher than the projected BMP values, indicating improved methane generation during the digestion process. The positive percentage increase suggests that co-digestion of PM and RSW at a 1:2 ratio enhanced biogas production efficiency. This indicates that this ratio provides better substrate synergy and microbial activity for methane generation.

❖ **Actual value of BMP and Targeted value of BMP of ratio 1:3 (PM: RSW)**

No. of Days	Actual BMP	Projected BMP	% Increase in BMP
	Nml CH ₄ /gm VS	Nml CH ₄ /gm VS	Nml CH ₄ /gm VS
1	130.3	135.2	-3.62
2	216.2	221.9	-2.57
3	267.6	271.0	-1.25
4	277.8	282.5	-1.66
5	284.2	288.8	-1.59
6	287.8	293.4	-1.91
7	290.9	296.7	-1.96
8	294.2	299.8	-1.87
9	297.2	302.5	-1.75
10	300.3	305.1	-1.57
11	303.5	307.7	-1.36
12	306.1	310.0	-1.26
13	308.4	312.1	-1.19
14	310.4	313.9	-1.12

Table 5.2.2: Actual & Targeted BMP values of ratio 1:3

The results indicate that the actual BMP values are slightly lower than the projected BMP values, resulting in a negative percentage increase throughout the digestion period. This suggests that the 1:3 ratio did not perform efficiently for methane production in this set.

5.3 Comparison of Co-digestion studies of both sets

Table VII. shows the total methane yield produced from different mixing ratios of both the substrates over 15 days by comparing two separate test groups (Set 1 and Set 2).

Sr. No.	Set	Samples	Moisture (%)	TS (%)	VS (%)	VS/TS (%)	15 th day BMP (N ml CH ₄ /g VS)	Projected BMP (N ml CH ₄ /g VS)	% increment wrt. Projected BMP
1	Set 1	RSW	85.49	14.51	10.93	75.33%	258.9		
2		PM	71.59	28.41	20.20	71.11%	334.7		
3		PM:RSW 1:2	80.85	19.15	14.02	73.24%	289.8	284.2	2.0
4		PM:RSW 1:3	82.01	17.99	13.25	73.66%	281.5	277.9	1.3
1	Set 2	RSW	86.78	13.22	9.92	75.04%	300.2		
2		PM	73.00	27.00	20.19	74.78%	362.9		
3		PM:RSW 1:2	82.19	17.81	13.34	74.90%	323.3	321.1	0.7
4		PM:RSW 1:3	83.35	16.66	12.49	74.97%	312.6	315.9	-1.0

Table 4.3: Comparison of BMP of Co-digestion studies

The co-digestion studies of PM and RSW revealed that the 1:2 ratio exhibited a higher percentage improvement compared to the 1:3 ratio across both experimental sets. Furthermore, the experimental BMP values were found to be higher than the projected values in three out of four cases, indicating that co-digestion did not result in any adverse effects on biogas potential. The BMP of co-digestion showed only a marginal increase compared to the projected values, indicating that co-digestion had no significant negative effect but also did not provide substantial improvement. Additionally, for all

cases involving raw spent wash, press mud, and their co-digestion, approximately 90% of the BMP was achieved within the initial 7 days, demonstrating rapid biodegradability and efficient methane generation.

5.4 Assessment of Fertilizer Reduction using Sugar Industry By-Products

These theoretical calculations were conducted to assess the agricultural co-benefits of utilizing biomethanated spent wash and press mud, complementing the energy recovery potential. The study is based on real-time inputs from local farmers.

5.4.1 Fertilizer Reduction Potential using Biomethanated Spent Wash

The impact of applying Biomethanated Spent Wash (BMSW) on the consumption of chemical fertilizers in sugarcane production, evaluated per acre of cultivated land and expressed in kilograms per metric tonne (MT) of sugarcane yield. Under standard cultivation without BMSW, the net fertilizer requirements are 4.44 kg/MT for DAP, 3.33 kg/MT for Potash, 0.44 kg/MT each for Magnesium Sulphate, Sulphur, and Humic acid, 4.44 kg/MT for Urea, 0.01 kg/MT each for Calcium Nitrate and Boron, and 1.11 kg/MT for Ammonium Sulphate. With the application of BMSW, these requirements are reduced to 4.16 kg/MT for DAP, 3.17 kg/MT for Potash, 0.42 kg/MT each for Magnesium Sulphate, Sulphur, and Humic acid, 4.18 kg/MT for Urea, 0.01 kg/MT each for Calcium Nitrate and Boron, and 1.06 kg/MT for Ammonium Sulphate. This translates to a percentage reduction in fertilizer consumption of 6.4% for DAP, 4.8% for Potash, 4.8% for Magnesium Sulphate, 4.8% for Sulphur, 4.8% for Humic acid, 6.1% for Urea, 4.8% for Calcium Nitrate, 4.8% for Boron, and 4.8% for Ammonium Sulphate. These results collectively demonstrate that BMSW application leads to a consistent reduction of 4.8% to 6.4% across all fertilizer inputs, with the most significant savings observed in the two major nitrogen sources, DAP and Urea. Thereby affirming BMSW's role as an effective organic supplement that enhances nutrient use efficiency and reduces dependence on chemical fertilizers in sugarcane cultivation.

Basis of this study based on the production of sugarcane from one acre of cultivated land.									
Summary (BMSW)									
Particulars	DAP	Potash	Magnesium Sulphate	Sulphur	Humic	Urea	Calcium Nitrate	Boron	Ammonium Sulphate
Net Fertilizer required for per MT of Sugarcane produced (Kg/MT)	4.44	3.33	0.44	0.44	0.44	4.44	0.01	0.01	1.11
Fertilizer required for per MT of Sugarcane produced with application of Biomethanated Spent Wash (Kg/MT)	4.16	3.17	0.42	0.42	0.42	4.18	0.01	0.01	1.06
Percentage reduction in fertilizer consumption (%)	6.4	4.8	4.8	4.8	4.8	6.1	4.8	4.8	4.8

Figure 5.3.1: Application of BMSW

5.4.2 Fertilizer Reduction Potential using Press Mud

To assess the agronomic benefit of Compost/Pressmud as an organic amendment, the fertilizer requirements for sugarcane production have been evaluated based on one acre of cultivated land, with all inputs quantified in kilograms per metric tonne (kg/MT) of sugarcane produced, both in the presence and absence of Compost/Pressmud application. Under standard cultivation without Compost/Pressmud, the net fertilizer requirements are 4.44 kg/MT for DAP, 3.33 kg/MT for Potash, 0.44 kg/MT each for Magnesium Sulphate, Sulphur, and Humic acid, 4.44 kg/MT for Urea, 0.01 kg/MT each for Calcium Nitrate and Boron, and 1.11 kg/MT for Ammonium Sulphate. With the application of Compost/Pressmud, these requirements are reduced to 3.57 kg/MT for DAP, 2.90 kg/MT for Potash, 0.39 kg/MT each for Magnesium Sulphate, Sulphur, and Humic acid, 3.76 kg/MT for Urea, 0.01 kg/MT each for Calcium Nitrate and Boron, and 0.97 kg/MT for Ammonium Sulphate. This translates to a percentage reduction in fertilizer consumption of 19.8% for DAP, 13.0% for Potash, 13.0% for Magnesium Sulphate, 13.0% for Sulphur, 13.0% for Humic acid, 15.3% for Urea, 13.0% for Calcium Nitrate, 13.0% for Boron, and 13.0% for Ammonium Sulphate. These results collectively demonstrate that Compost/Pressmud application leads to a substantially higher reduction of 13.0% to 19.8% across all fertilizer inputs compared to conventional practices, with the most significant savings observed in DAP (19.8%) and Urea (15.3%), thereby affirming that Compost/Pressmud serves as a highly effective organic amendment that considerably enhances nutrient use efficiency and significantly reduces dependence on chemical fertilizers in sugarcane cultivation.

Summary (Compost/Pressmud)									
Particulars	DAP	Potash	Magnesium Sulphate	Sulphur	Humic	Urea	Calcium Nitrate	Boron	Ammonium Sulphate
Net Fertilizer required for per MT of Sugarcane produced (Kg/MT)	4.44	3.33	0.44	0.44	0.44	4.44	0.01	0.01	1.11
Fertilizer required for per MT of Sugarcane produced with application of Compost (Kg/MT)	3.57	2.90	0.39	0.39	0.39	3.76	0.01	0.01	0.97
Percentage reduction in fertilizer consumption (%)	19.8	13.0	13.0	13.0	13.0	15.3	13.0	13.0	13.0

Figure 5.3.2: Application of Press Mud

Chapter 6
CONCLUSION AND FUTURE SCOPE

6. CONCLUSIONS

6.1 Conclusion

This research presents a comprehensive feasibility analysis of the optimal mixing ratio of press mud (PM) and raw spent wash (RSW) for enhanced biogas production using Biochemical Methane Potential (BMP) analysis. Detailed characterization of both substrates was carried out to evaluate their suitability for anaerobic digestion. The study aimed to assess the biodegradability of PM and RSW, understand their individual and combined methane generation potential, and determine the effectiveness of co-digestion in improving process stability and overall biogas yield.

The major findings of the present research are summarised as follows:

1. Raw spent wash (RSW) with a high COD (~140,000 mg/L), exhibited with BMP values of 258.9 N ml CH₄/g VS (Set 1) and 300.2 N ml CH₄/g VS (Set 2), indicating moderate methane potential compared to press mud.
2. Press mud (PM), when used as a sole substrate, achieved the BMP of 334.7 N ml CH₄/g VS (Set 1) and 362.9 N ml CH₄/g VS (Set 2), confirming its strong biodegradability.
3. The study evaluated the feasibility of co-digesting press mud (PM) with raw spent wash (RSW) to enhance methane production and process performance.
4. Co-digestion (PM: RSW 1:2) resulted in 289.8 N ml CH₄/g VS (Set 1) and 323.3 N ml CH₄/g VS (Set 2), which were higher than RSW alone, indicating improved methane generation through co-digestion.
5. Co-digestion (PM: RSW 1:3) yielded 281.5 N ml CH₄/g VS (Set 1) and 312.6 N ml CH₄/g VS (Set 2), also showing better performance than mono-digestion of RSW.
6. The co-digestion BMP values were very close to the expected values, with only small differences (−1% to +2%), showing no major improvement due to synergy.
7. In all raw spent wash, press mud, and co-digestion cases, 90% of BMP was achieved within the first 7 days, indicating rapid biodegradation.
8. Among the mixing ratios, PM: RSW (1:2) performed slightly better than PM: RSW (1:3) in both sets.

9. The combination of PM and RSW helped in maintaining a more stable pH environment, where PM provided buffering capacity and prevented acidification from high-strength RSW.
10. Overall, co-digestion of press mud and raw spent wash was found to be more efficient than RSW alone and operationally stable, but did not significantly enhance methane production compared to mono-digestion of press mud.

6.2 Future Scope and Recommendations

Although the present study provides valuable insights into the bio-methanation potential of press mud and raw spent wash, further research can be carried out to improve the efficiency and applicability of the process.

- a) Pilot-scale and industrial-scale studies can be conducted to evaluate the feasibility of large-scale anaerobic digestion systems for press mud and spent wash.
- b) Further research can focus on optimizing different mixing ratios and operational parameters such as temperature, pH, hydraulic retention time, and organic loading rate to maximize methane yield.
- c) The application of pre-treatment techniques for lignocellulosic materials in press mud may enhance biodegradability and improve biogas production.
- d) Additional studies can investigate the co-digestion of press mud and spent wash with other organic wastes such as cow dung, food waste, or agricultural residues to further increase methane yield.

6.3 Societal Importance

1. Sustainable Management of Industrial and Agro-Waste:

The sugar and distillery industries generate large quantities of waste such as press mud and raw spent wash, which pose disposal challenges. This study promotes their effective utilization through anaerobic co-digestion. It ensures environmentally sound waste management practices and minimizes ecological degradation.

2. Renewable Energy Generation through Waste Valorisation:

The utilisation of press mud and raw spent wash for extracting biogas provides a sustainable source of renewable energy. Methane-rich biogas can be used for

electricity generation, heating, or as a substitute for fossil fuels. This reduces reliance on non-renewable energy sources and supports India's renewable energy goals.

3. Improvement in Agricultural Resource Efficiency:

The digestate produced after biomethanation is rich in nutrients such as nitrogen, phosphorus, and potassium. It can be used as an organic fertilizer, reducing dependence on chemical fertilizers. This enhances soil fertility, promotes sustainable agriculture, and improves crop productivity. It also creates employment opportunities in rural areas for plant operation, maintenance, and waste collection, thereby improving local livelihoods.

4. Promotion of Circular Economy in Sugar and Distillery:

This approach supports a circular economy by converting waste into valuable resources like energy and fertilizer. It ensures efficient resource utilization, reduces waste generation, and promotes sustainable industrial practices.

Chapter 7
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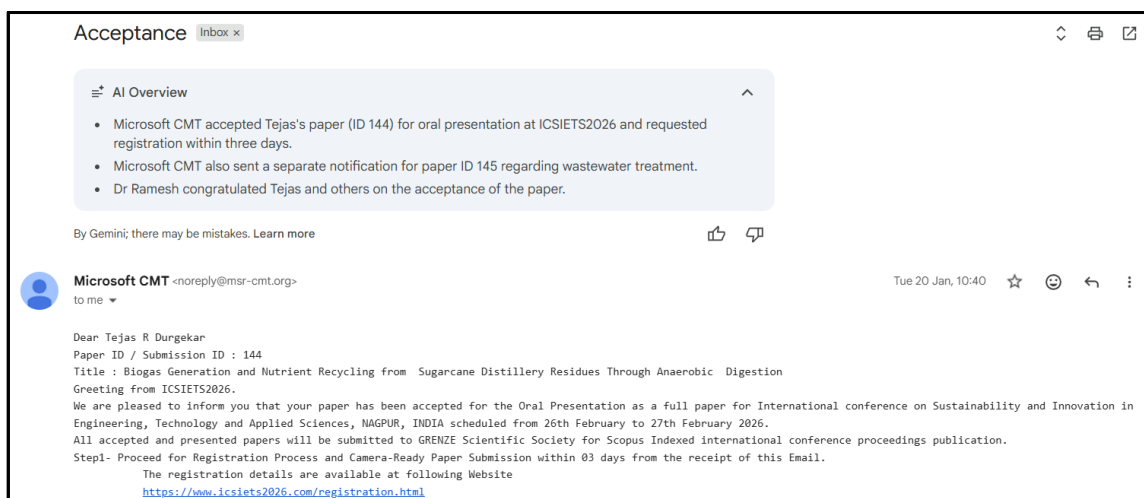
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